Astana IT University
School of Software Engineering

Pre-defense 1 report

Development of a Machine Learning-Based System for Analyzing Patient Health Indicators

Design of an early sepsis-warning system for intensive care units

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Abstract

This report presents the analysis and design of a system that estimates the risk of sepsis for patients in an intensive care unit (ICU) from their hourly vital signs and laboratory results, and that delivers a warning to the nurse together with the reasons for it. The work is approached from the software-engineering side: the main contribution is a system that can be followed from the hospital's data to the nurse's response, with every step specified, justified and testable. The machine-learning model is one component of it and is evaluated against two simple baselines and on data from a second hospital system.

The report describes how ICU patients are monitored and how deterioration is escalated in Kazakhstan, based on the national ICU standard, the national sepsis protocol and two interviews with clinical staff. It then reviews twelve studies and four deployed systems, derives 19 functional requirements (with two more explicitly excluded) and 10 non-functional requirements, and presents the architecture of the target system in a hospital and of the version built in this diploma, which replays the open PhysioNet/CinC Challenge 2019 data hour by hour through the same data-source interface. A clickable prototype, the API contract, the verification and evaluation plan and the project plan complete the pre-defense material. The system is a research prototype; it does not diagnose and is not intended for clinical use.

Keywords: sepsis, early warning, intensive care, clinical decision support, software architecture, machine learning, PhysioNet.

Introduction

Sepsis is a life-threatening organ dysfunction caused by a dysregulated response to infection [1]. The Global Burden of Disease study estimated 48.9 million cases and 11 million sepsis-related deaths in 2017, about one in five deaths worldwide [2], and the World Health Organization stresses that treatment is most effective when it starts early [3]. The Surviving Sepsis Campaign guidelines therefore recommend that hospitals run screening programmes. They also advise against using qSOFA alone as a screening tool and note that machine learning may improve screening [4].

Early recognition depends on routine measurements. The Sepsis-3 definition identifies organ dysfunction with the SOFA score, which is built largely from vital signs and laboratory values [1]. In an ICU these values are recorded every hour, but they are spread over the bedside monitor, the nurse's chart and the laboratory system, and many of them are missing at any given hour. A nurse who looks after two patients through a night shift has to notice a slow combination of changes, such as a rising heart rate, faster breathing and a falling blood pressure, before any single value is alarming.

In Kazakhstan, intensive care is organised by a national standard [5], and the treatment of sepsis follows a national clinical protocol [6]. The protocol defines what the physician does once sepsis is suspected, with a clear clock for lactate, blood cultures and treatment. It does not define how the suspicion arises. The two clinicians interviewed for this project described the same chain: a nurse notices a change during a routine measurement, checks the patient and informs the physician. They also described where it breaks: early signs are taken as minor, changes are lost at shift handover and at night, and one nurse has many patients. One of them summarised it as the absence of a system of red flags (Appendix A, R2-Q7). I did not find a national estimate of sepsis incidence for Kazakhstan, but a study of a 12-bed ICU in Astana describes hospital-associated infections there, including bloodstream infections and ventilator-associated pneumonia, which can lead to sepsis [7].

Systems that automate this early step exist abroad. The Epic Sepsis Model and the Epic Deterioration Index run inside a commercial health record, COMPOSER and CHARTwatch were built by individual hospitals, and TREWS is a commercial product. Their published results are mixed: some deployments were followed by lower mortality [8, 9], while an independent validation of the most widely used model found that it missed two thirds of sepsis cases [10]. All of them are proprietary or in-house, and their publications describe the model in more detail than the software and the workflow around it.

The problem addressed in this diploma is therefore an engineering one. There is no open, tested system that takes hourly ICU measurements, estimates the risk of sepsis in a way that can be inspected, compares that estimate fairly with simple clinical rules and on a second hospital system, and delivers the warning into the nurse's existing workflow with a defined response. The project designs such a system for the setting of a Kazakhstani ICU and builds it on open data.

Aim. To design, implement and evaluate a reproducible decision-support system that analyses hourly ICU measurements, estimates the risk of sepsis about six hours before onset, and presents each warning with its reasons and a defined response path.

Objectives.

1) Describe how ICU patients are monitored and how deterioration is escalated in Kazakhstan, and define the place of the system in that process.
2) Define a reproducible early-prediction task on the open PhysioNet/CinC Challenge 2019 data.

3) Design a data-source interface that lets the same system run on replayed open data and, in a hospital, on data from the medical information system.
4) Implement validation and feature building that never use future values.
5) Train a gradient-boosted model and compare it with a partial SIRS rule and logistic regression, internally and on a second hospital system.
6) Implement the alert rules, the acknowledgement of alerts by the nurse, storage with an audit trail, a REST API and a dashboard.
7) Cover the critical behaviour with automated tests and document how every result can be reproduced.

Research questions. The work answers three questions. The first (RQ1) is how well a gradient-boosted model that uses only the data recorded up to the current hour predicts sepsis about six hours in advance, compared with a partial SIRS rule and a logistic regression. The hypothesis is that LightGBM reaches a higher area under the precision–recall curve (AUPRC) than logistic regression on the internal test set, with a 95% confidence interval of the paired difference that excludes zero. The second (RQ2) is how much performance changes on patients from a different hospital system; all models are expected to lose performance, and the size of the drop is reported for each. The third (RQ3) is whether potential users interpret the warning correctly: at least two of three respondents should explain from the patient screen why the risk is high.

Focus of the work. At the meeting on 23 October 2026 the supervisor asked for a choice between a very well-developed system, seen as a software engineer, and a very deep machine-learning methodology. I chose the first, which matches my major. The model is trained and evaluated with care, but the depth of the work lies in the design of the system, its fit with the clinical workflow and its verification. For the same reason the report separates two things throughout: the **target system**, meaning the system as it would work in a hospital, and the **diploma implementation**, meaning the part that is built and tested on open data.

Scope. The diploma includes data validation, leakage-safe features, the baselines and the main model, internal and external evaluation, alert logic, explanations, acknowledgement and audit, a REST API, a dashboard, tests and documentation. It excludes diagnosis or treatment advice, use on real patients, a working connection to a hospital information system (it is designed but not built) and clinical validation. The system uses only public, de-identified data and runs on one laptop.

Structure of the report. Chapter 1 presents the evidence that the problem is real: two interviews and an analysis of the dataset. Chapter 2 reviews the literature and Chapter 3 the existing systems and how they are built. Chapter 4 describes intensive care in Kazakhstan and the place of the system in it. Chapter 5 justifies the technology and the machine-learning method, Chapter 6 states the requirements, and Chapter 7 presents the design of the system. Chapter 8 describes the prototype, Chapter 9 the verification and evaluation plan, Chapter 10 the project management and risks, and Chapter 11 the supervisor's feedback and how it was addressed. Appendix A contains the interview transcripts and Appendix B the traceability from requirements to tests.

1 Relevance of the problem and user evidence

The relevance of the problem was checked in two ways: by asking people who work with patients how deterioration is noticed today, and by analysing the dataset to see whether it has the properties that the problem statement assumes.

1.1 How the evidence was collected

On 21 October 2026 an anonymous survey (Google Forms, no login and no e-mail collection) was sent to eight to ten people with clinical experience. It had thirteen questions in English and Russian about how deterioration is noticed, where the process fails, which scores and alerts are used, and what a risk tool would have to show to be trusted. Two people answered, both in Russian, and both left a contact for follow-up questions. Each of them then answered five further questions in a messenger; the questions were written for each respondent from their survey answers.

The first respondent (R1) is a nurse on a general medical or surgical ward with more than five years of experience. The second (R2) is a resident with one to three years of experience in an ICU, a urology–nephrology department and the operating theatre. Their full answers are given in Appendix A in the original Russian and in translation. The survey consent covered anonymous use in summarised form; after the supervisor asked for transcripts, both respondents were asked again and agreed to the publication of their anonymised answers in full. Names and contact details are not stored in the repository.

Two respondents are below the target of three to five discussions, and neither answer can be taken as representative of all ICUs in Kazakhstan. The evidence is used to check assumptions and to shape the design, not to make claims about practice in general. Further respondents and a user check of the prototype screens are planned (Chapter 9).

1.2 What the respondents said

Both respondents describe the same escalation path. A nurse notices a change while measuring blood pressure or pulse, looks at the patient, checks consciousness and vital signs, and informs the physician; in a critical situation the nurse starts the necessary measures within their competence (R1-Q6, R2-Q6). The schedule of measurements depends on the department: "ICU — every hour; ward patients — every 4 or 8 hours; stable patients — every 12 hours", and the values are read "mostly on the monitor or in the prescription sheet" (R2-Q5).

The respondents locate the weak points at the beginning of this path. According to R1, failures happen "at the stage of detecting deterioration and when information is passed between staff": fast breathing or unusual drowsiness are taken as minor changes, such changes go unnoticed at a shift change, and a doctor may be unavailable (R1-Q7). R2 names a large number of patients and night time as the reasons why deterioration was noticed late (R2-FU4) and writes about the absence of a red-flag system apart from code blue (R2-Q7). These answers support the assumption that early detection of deterioration is a real difficulty and that the nurse is the right primary user.

The answers about alerts differ. R1 has worked with automatic alerts in a hospital information system and says that their usefulness depends on accurate measurements, timely data entry and the staff's response; false alarms and high workload cause alerts to be lost (R1-Q9, R1-Q10). R2 states that "alerts are not ignored" (R2-Q10). The concern about alert fatigue is therefore supported by one respondent and by the literature (Chapter 2), but not by the other.

Both respondents were clear that a number alone is not enough. R1 wants to see "which measurements changed, when they were measured and what exactly is of concern" (R1-Q11); R2 wants the clinical picture and the trend next to the risk (R2-Q11). From a list of possible features, both chose the measurements that influenced the risk, a warning when data are missing or out of date, and a list of patients sorted by risk; R2 also chose charts over time and a limit on repeated alerts (Q12). Both insisted that the system must not replace professional judgement (R1, final question; R2-Q13).

The follow-up questions showed three disagreements that matter for the design. On how old data may be, R1 would flag measurements not updated for four to six hours on a ward, while R2 considers ICU data out of date if they are not updated every hour. On how to show the risk, R2 prefers a percentage and R1 a colour with a short label such as "High risk" in red, because it is faster to read under workload. On repeated alerts, R1 does not want identical notifications all the time but wants a repeat when the state changes or the situation is dangerous, and R2 wants an alert at every change (R1-FU1–FU3, R2-FU1–FU3).

These answers changed the design in five places, all adopted on 23 October 2026. The risk is shown as a percentage together with a coloured label, so that both readings are available. A new alert is raised inside the suppression window when the risk rises further. The time since the last measurement is shown next to each vital sign. The limit for "insufficient data" is a setting with a default of four hours, so that an ICU can choose one hour. Every alert carries one fixed prompt, "Re-check the measurements and inform the responsible physician", which is what R1 asked for when she described the help she would want (R1-FU5). Treatment advice stays outside the scope.

1.3 What the data show

The full public training data of the PhysioNet/CinC Challenge 2019 [11, 12] were analysed with a reproducible script (`make eda`). The data come from two hospital systems, called set A and set B, and contain one row per patient-hour with up to 40 variables: vital signs, laboratory values, demographic data and the label. Table 1.1 gives the main figures.

Table 1.1 – Main properties of the PhysioNet/CinC Challenge 2019 training data

Property	Set A	Set B	Both
ICU stays	20,336	20,000	40,336
Stays with sepsis	1,790 (8.80%)	1,142 (5.71%)	2,932 (7.27%)
Patient-hours with SepsisLabel = 1	2.17%	1.42%	1.80%
Septic stays positive from the first hour	203	223	426
Length of stay, median (IQR), hours	39 (25–47)	38 (23–47)	38 (24–47)

The analysis supports the problem statement in four respects. First, the data are irregular and incomplete in the way the introduction describes. Heart rate is missing in 7.7–12.1% of hours, but each of the 26 laboratory values is missing in 83–99.8% of hours. Second, sepsis is present but rare at the level of hours (1.8%), so a model that predicts "no sepsis" for every hour would look accurate. This is why the evaluation reports AUPRC and alert metrics next to AUROC. Third, there is time to warn. Leaving aside the 426 stays that are positive from the first hour, the derived onset has a median of 43 hours after the first record (IQR 19–89 hours). Fourth, the two hospital systems differ. Set B has fewer septic stays, BaseExcess and HCO3 are almost never recorded in it, and EtCO2 is never recorded in set A. This supports an external test on set B and explains part of any drop in performance there.

The analysis also gave a data-based value for one requirement. The six key vital signs are measured almost every hour: only 0.23% of the gaps between their measurements are longer than four hours. A gap of more than four hours is therefore unusual and is a reasonable default for the "insufficient data" state, which agrees with R1's answer of four to six hours.

1.4 Status of the assumptions

Before the evidence was collected, the project rested on six assumptions. The evidence supports four of them: that ICU nurses and physicians are the right primary users, that trends of repeated measurements are useful next to a risk, that a warning needs supporting context, and that a risk value alone is not enough to act on. It also supports the assumption that early detection of deterioration is a suitable concrete problem. The assumption that too many alerts reduce usefulness is only partly supported: R1 and the literature support it, R2 does not. The laboratory data are confirmed to be missing in most hours. What remains open is whether the proposed alert path, the 30-minute limit for acknowledgement and the size of the re-alert margin match practice in an ICU; these are the questions for the next discussions (Section 9.4).

2 Literature review

The review asks what is known about predicting sepsis and clinical deterioration from routine measurements, how such predictions are evaluated, and what happens when they are used in hospitals. Its purpose is to find the limitations that a new system should respond to.

2.1 Search and selection

The first six sources were found on 10 October 2026 in PubMed/PMC and by web search, with queries combining sepsis, early prediction, machine learning, intensive care and early warning scores. Six more were added on 13 October to cover external validation, uncertainty and deployment. A source was kept if it informed the prediction task, the evaluation, the validation on new data or the software around the model. All twelve full texts were read. Eleven are from 2021–2024; the paper that defines the dataset is from 2020 and is kept for that reason. The queries, dates and excluded candidates are logged in the repository.

2.2 Machine learning and simple scores

Machine-learning models usually discriminate better than simple scores, but not always, and not by as much as single studies suggest. Lashen et al. found that gradient boosting (AUROC 0.778) predicted deterioration better than the National Early Warning Score (NEWS, 0.682), while logistic regression (0.654) did not [13]. A network meta-analysis of 73 studies found a pooled AUC of 0.825 for machine-learning models against 0.719 for NEWS and NEWS2, with high heterogeneity and a high risk of bias in many studies [14]. In a comparison across seven hospitals, NEWS (AUROC 0.829) outperformed some proprietary AI scores, including the Epic Deterioration Index (0.808) [15]. The conclusion for this project is that a model must be compared with transparent baselines on the same patients, not assumed to be better.

2.3 Performance at a new hospital

Performance measured where a model was developed often does not hold elsewhere. In the PhysioNet/CinC Challenge 2019, 104 teams submitted 853 entries, and the ranking of algorithms changed on a hospital system whose data the teams had never seen [11]. An independent validation of the Epic Sepsis Model found an AUC of 0.63; at the threshold in use it missed 67% of sepsis cases and alerted in 18% of hospitalisations [10]. Moor et al. trained deep models on four ICU databases and found that the mean internal AUC of 0.846 fell to 0.761 at new sites; at 80% sensitivity the positive predictive value (PPV) was 42%, about 1.4 false alerts for each true one [16]. The project therefore keeps set B of the Challenge data as an external test that is never used for training or tuning.

2.4 What has to be measured

A systematic review of 22 studies of sepsis prediction in the ICU found that definitions of sepsis, prediction horizons and evaluation methods differ so much that results cannot be compared [17]. The area under the ROC curve alone hides poor performance on a rare outcome and says nothing about how many alerts the staff will receive. The Challenge organisers showed that AUROC and a utility score that rewards early and penalises late predictions can rank algorithms differently [11]. Shashikumar et al. added a conformal layer to the COMPOSER model, so that it answers "indeterminate" when the input does not resemble its training data; this reduced false alarms on external data by 78–86% [18]. For this project it follows that the label and horizon must be fixed and documented, that AUPRC, PPV at a fixed sensitivity, calibration and alert frequency

must be reported next to AUROC, and that the system should be allowed to say that it does not have enough data.

2.5 The workflow decides the benefit

The studies that report benefits for patients did not test a model alone. COMPOSER was deployed in two emergency departments with an alert to the nurse inside the health record; a quasi-experimental study found a 17% relative reduction in in-hospital mortality [8]. CHARTwatch was introduced on an internal medicine unit with alerts to the team and a defined care pathway; non-palliative deaths fell from 2.1% to 1.6% (adjusted relative risk 0.74) in a non-randomised evaluation [9]. At one hospital the Epic Deterioration Index was combined with an alert and a structured huddle at a score of 65, which reduced a composite outcome by 10.4 percentage points near the threshold [19]. A review of 37 implementation articles found that performance falls from retrospective tests to live use, that thresholds and alert suppression are common, and that relatively simple models can support successful implementations [20]. In every case the benefit came from a package of model, threshold, alert and human response. A student project cannot test clinical outcomes, but it can design the whole package and make no claim beyond what it tests.

2.6 The gap this project addresses

The gap is not that models for early sepsis prediction do not exist. Published models are hard to compare, lose performance at new hospitals and can produce many false alerts, and their usefulness depends on data quality, transparency and the human response around them. The deployed systems that address these issues are proprietary or in-house. The project responds with an open system that applies time-valid preprocessing to open data, compares the model with reproducible baselines internally and on a second hospital system, reports alert-aware metrics, has an explicit state for insufficient data, and specifies the clinical response to each alert. Table 2.1 summarises how each limitation is addressed.

Table 2.1 – Limitations found in the literature and the response of the project

Limitation in the literature	Response in this project
Definitions and horizons differ, so results cannot be compared [17]	One documented label and horizon for all models
Performance drops at a new hospital [11, 10, 16]	External test on set B, never used for training or tuning
AUROC hides poor performance on rare events and the alert burden [17, 10]	AUPRC, PPV at fixed sensitivity, calibration, alerts per patient-day
Machine learning does not always beat simple scores [13, 14, 15]	Partial SIRS rule and logistic regression as baselines, paired bootstrap comparison
Confident output on unfamiliar or stale data [18]	"Insufficient data" state instead of a risk value
Benefit depends on the workflow [8, 9, 19, 20]	Alert path, acknowledgement and introduction plan designed; no claims of clinical benefit

3 Existing systems and how they are built

Five systems were selected that watch repeated patient measurements and warn clinicians about sepsis or deterioration, and that have a peer-reviewed evaluation: the Epic Sepsis Model, the Epic Deterioration Index, COMPOSER, CHARTwatch and TREWS. They were compared on the same nine criteria, from the problem and method to the evidence and access; the full comparison is in the repository. This chapter adds what the supervisor asked for on 23 October: how each system is built.

None of the five systems has a published entity–relationship diagram or an official architecture diagram. For COMPOSER, CHARTwatch and TREWS, the publications describe the data flow, the deployment and the way alerts reach the staff in enough detail to reconstruct a simplified architecture. For the two Epic models the vendor has published nothing about the internals, and only independent studies describe their inputs, thresholds and use. The figures in this chapter are therefore my reconstructions. Every element in them comes from the cited text, and where a source does not describe something, the figure says so.

3.1 COMPOSER

COMPOSER, developed at UC San Diego Health, is the most fully described deployment [8, 18]. The model does not run inside the health record. It runs on a cloud platform, and the hospital's Epic record is connected to it in three ways. An admission–discharge–transfer (ADT) feed tells the platform which patients are in the emergency department. Every hour the platform pulls their clinical data through the record's FHIR interface, authorised with OAuth 2.0. The score is written back into the record as an HL7 v2 message. When the score passes the threshold, the nurse sees a Best Practice Advisory, an alert window inside Epic that lists the features that contributed most to the score. One of the options in it is to notify the physician at once, and in the study that choice was associated with a shorter time to antibiotics [8]. The deployment also monitors data quality and model performance, and the model can answer "indeterminate" when the input does not resemble its training data [18]. Figure 3.1 shows the reconstruction.

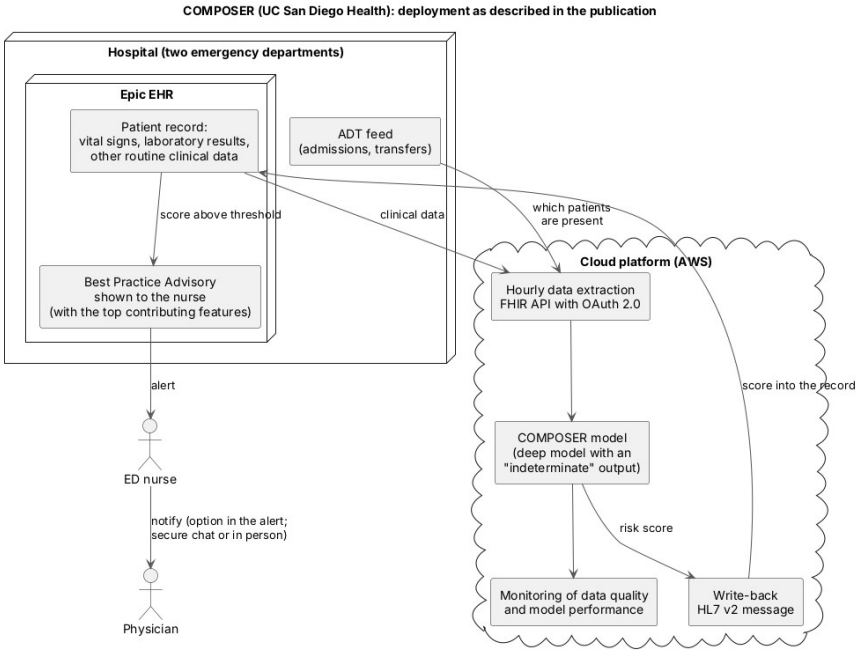


Figure 3.1 – COMPOSER: deployment reconstructed from the publications

Three ideas are taken from COMPOSER. Getting data from the hospital, computing the risk and showing the result inside the staff's normal tools are separate concerns; in this project the data-source adapter plays the role of the FHIR extraction. The nurse is the first recipient, and the physician is reached through a choice made in the alert, which becomes the "physician informed" acknowledgement. The "indeterminate" output is the model for the "insufficient data" state.

3.2 CHARTwatch

CHARTwatch, built at St. Michael's Hospital in Toronto, shows how far a relatively simple model can go with careful engineering around it [21, 9]. The model was trained on historical data from the hospital's enterprise data warehouse. In production, a pipeline written in R runs every hour on a local server. It extracts the current census and each patient's data from the hospital's source databases, cleans them, builds six-hour windows, scores them with a MARS model and assigns a high, medium or low risk group (Figure 3.2).

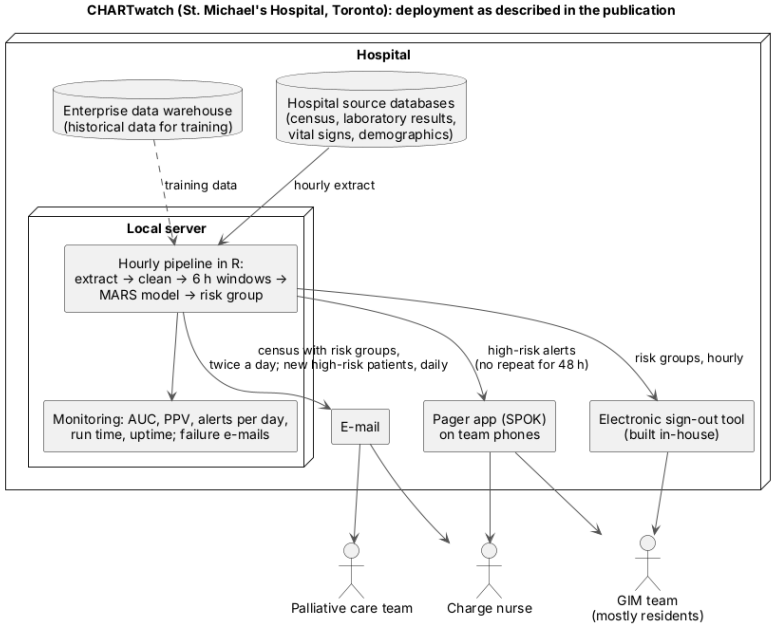


Figure 3.2 – CHARTwatch: deployment reconstructed from the publications

The results reach people through tools they already use. High-risk alerts go to a pager application on the phones of the medical team and the charge nurse. The risk groups appear every hour in an electronic sign-out tool built in the hospital. E-mail sends the charge nurse a census with risk groups twice a day, for assigning nurses, and sends the palliative care team a daily list of new high-risk patients. The alert rules are strict: no repeat for the same patient for 48 hours, a new alert only after the patient has dropped to medium or low risk and returned to high, silence for 24 hours after a transfer from the ICU, and, since 2022, no more than five alerts per patient. Before going live the system ran in a silent period. The team monitors AUROC, PPV, alerts per day, run time and uptime, failures trigger e-mails, and an implementation committee reviewed the figures first weekly and then monthly. Over 14,016 hours the system was available 99.6% of the time [21].

From CHARTwatch the project takes the suppression of repeated alerts with an exception, the use of channels the staff already use, and the introduction through a silent period with monitoring of alert frequency and uptime. The suppression window here is shorter (six hours rather than 48), because both respondents asked for a new alert when the patient's state changes.

3.3 TREWS

TREWS was deployed in five hospitals in the United States and uses routine record data: laboratory results, vital signs, notes, and the history of medications and procedures [22, 23]. Its main design choice is a passive alert (Figure 3.3). Instead of a pop-up, the patient gets an icon in the patient list. Clicking it opens a TREWS page in the chart that shows the factors behind the alert, the estimated probabilities of sepsis and death, and the status of sepsis treatment. Nurses can pre-screen the alert and pass it to a physician or advanced practice provider, who evaluates it and confirms or dismisses it; an evaluation within three hours counts as timely. In the multi-site study, patients whose alert was confirmed within three hours had lower adjusted mortality [23]. The sources do not say where the model is computed or how often.

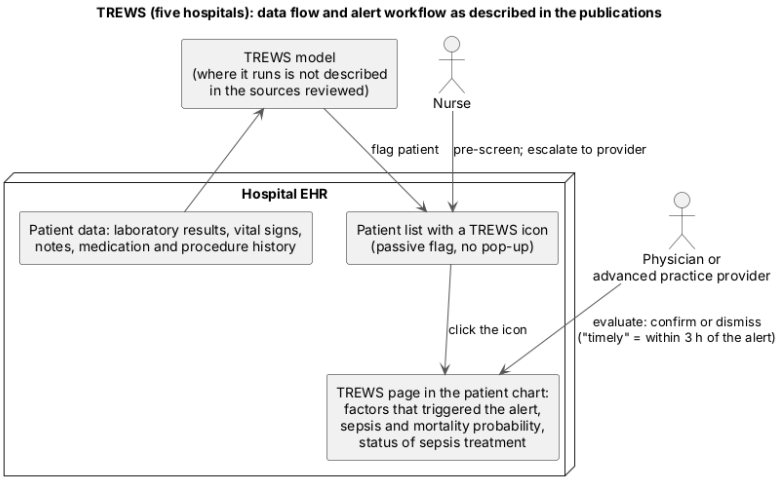


Figure 3.3 – TREWS: alert path reconstructed from the publications

TREWS shows that a flag in the patient list with a short page of reasons interrupts less than a pop-up, and that recording the clinician's response makes the alert measurable in use. The patient list with alerts at the top, the patient view with five reasons and the acknowledgement with an outcome in this project follow the same idea.

3.4 Epic Sepsis Model and Epic Deterioration Index

Both models run inside the Epic health record and belong to the vendor; their architecture, data model and update schedule are not published (Figure 3.4). Independent studies describe them only from the outside. The Sepsis Model is a penalised logistic regression; at the threshold in use it found a third of sepsis cases and alerted in 18% of hospitalisations, with an external AUC of 0.63 [10]. The Deterioration Index is an ordinal logistic regression over 31 measures, updated every 15 minutes. At the study site a score of 65 sent an alert to the nurse and the physician and started a structured huddle with a checklist [19]. In the seven-hospital comparison its AUROC was lower than that of NEWS [15].

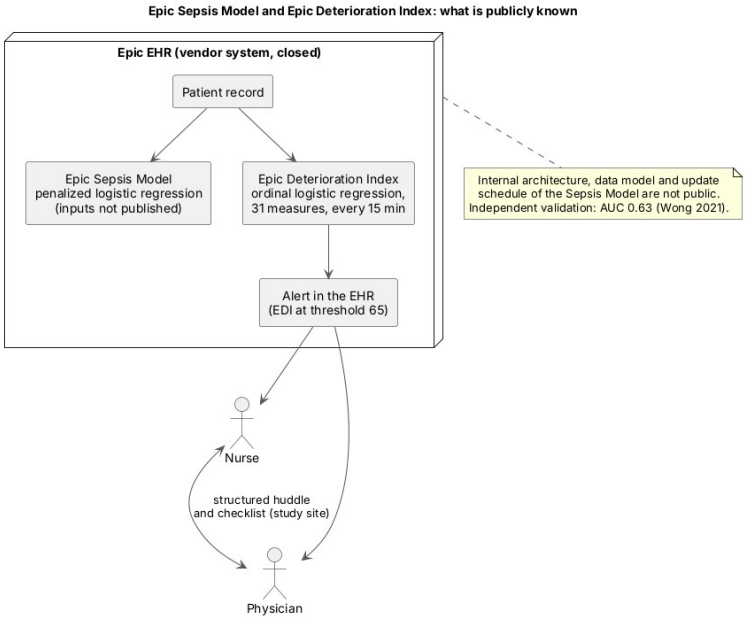


Figure 3.4 – Epic Sepsis Model and Deterioration Index: what is publicly known

A closed model cannot be inspected, and its published performance may not hold in a new hospital. This is the strongest argument for the openness of the system proposed here: open code, a model card, an external test on a second hospital system, and a comparison with a simple score.

3.5 Comparison

Table 3.1 condenses the comparison. All five systems are proprietary or built in-house; none publishes its software. Only COMPOSER shows the user the features behind a warning, and only COMPOSER and CHARTwatch describe how the system is monitored after go-live.

Table 3.1 – Comparison of the existing systems

	Epic Sepsis Model	**Epic Deterioration Index**	**COMPOSER**	**CHARTwatch**	**TREWS**
Target	Sepsis, inpatients	Deterioration, wards	Sepsis, ICU and ED	Death or ICU transfer, internal medicine	Sepsis
Method	Penalised logistic regression	Ordinal logistic regression, 31 measures	Deep model with "indeterminate" output	MARS on 6-hour windows	Not reported
Where it runs	Inside the vendor's record	Inside the vendor's record	Cloud platform, FHIR in, HL7 out	Local server, hourly R pipeline	Not reported
Alert delivery	In the record	Nurse and physician, huddle	Alert window to the nurse	Pager, sign-out tool, e-mail	Icon in the patient list, page of reasons
Explanation	Not reported	Not reported	Top features	Not reported	Factors behind the alert
Evidence	External AUC 0.63	−10.4 pp composite outcome	17% relative mortality reduction	Death 2.1% → 1.6%	Lower mortality with timely confirmation
Access	Proprietary	Proprietary	In-house	In-house	Commercial

The four publicly described architectures share a pattern. Data are taken from the hospital record on a schedule, the risk is computed outside or beside the record, the result is returned into tools the staff already use, the response of the staff is part of the design, and the system is monitored after go-live. They differ in where the model runs and in how hard they push the alert, from a pager message to a passive icon. The target system in Chapter 7 follows the same five steps and differs in three ways: the data source is a replaceable adapter, so that the same system runs on open data in the diploma and on a hospital information system in the target; the alert is shown at the nurse station and acknowledged with a recorded outcome; and the code, the model and the evaluation are open.

4 Intensive care in Kazakhstan and the place of the system

A system that warns about deterioration is only useful if it fits the way the unit works. This chapter describes how ICU patients are monitored in Kazakhstan, how deterioration is escalated, and where in that process the system belongs. It is based on three regulatory documents and on the two interviews. Practice differs between hospitals, so the chapter describes the common frame rather than one unit, and it will be checked with ICU staff in the next discussions.

4.1 Organisation and staffing

Intensive care is regulated by the Standard for organising anaesthesiology and resuscitation care, approved by Order No. 78 of the Minister of Health in 2023 and amended in 2025 [5]. An intensive care unit has at least six beds and a separate room for infectious and septic patients, and severe sepsis of any origin is one of the listed reasons for admission. The staffing norm is at least 4.75 nurse posts per two ICU beds. Since a round-the-clock post is commonly staffed with 4.75 positions, this corresponds to about one nurse on duty for every two beds; for patients on mechanical ventilation, ECMO and some other profiles the norm is one post per bed. The Standard requires continuous monitoring of the patient's state and a daily examination with the head of the department, and progress notes every three hours for unstable patients and every six hours for stable ones.

4.2 Measurements and where they are recorded

For anaesthesia, the Standard lists the monitored parameters: heart rate, ECG, oxygen saturation, blood pressure, temperature, ventilation and hourly urine output [5]. For ICU patients the respondents name the same bedside parameters together with the respiratory rate. R2 described a fixed recording schedule — every hour in the ICU, every four or eight hours on the ward, every twelve hours for stable patients — with the values read on the monitor or in the prescription sheet. Part of the information therefore exists only on the device until a nurse writes it down. Laboratory results come from the hospital laboratory: blood gases and acid–base balance, complete blood count, biochemistry and urinalysis (R2-Q4, R2-Q5).

All state medical organisations had medical information systems (MIS) by 2020, and the Ministry has been moving them to paperless records [24]. One MIS, Damumed, covered more than 80% of the market in 2023 [25]. The minimum requirements for medical digital systems were updated in July 2026 [26]. They allow integration with laboratory systems, diagnostic equipment and patient monitoring devices, and they name HL7 and DICOM as usable standards. They also require that software and data are hosted in Kazakhstan, together with role-based access, encryption and logging of user actions. A hospital-side early-warning system would therefore receive its data through the MIS or a device gateway, inside the hospital, under these rules.

4.3 How deterioration is recognised and escalated

Figure 4.1 shows the current process as the respondents describe it. The nurse charts the values from the monitor, watches several patients and, on noticing a change, checks the patient and informs the physician. The physician then follows the national clinical protocol "Sepsis" [6]. The protocol uses the Sepsis-3 definition, an acute rise of the SOFA score by at least two points in a patient with suspected infection, and qSOFA for screening outside the ICU. It sets a clear clock: lactate and blood cultures within 1.5 hours of a presumed diagnosis, fluid resuscitation within the first hour, antibiotics without delay, and control of the source of infection within six hours.

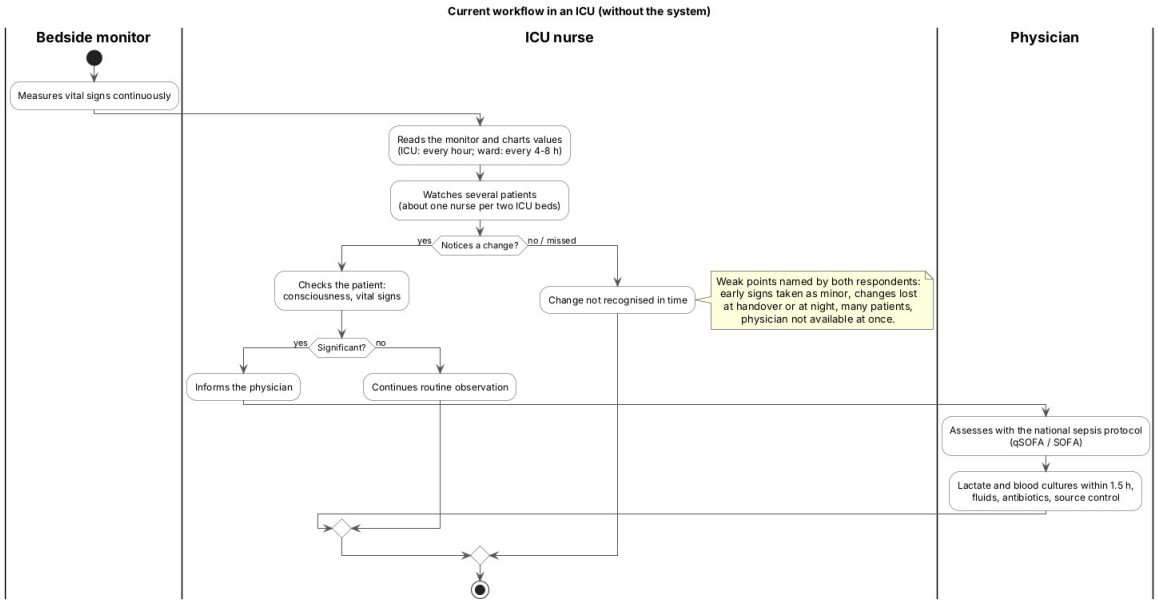


Figure 4.1 – Current workflow in an ICU, without the system

The weak points named by the respondents all lie before this clock starts: early signs are taken as minor, changes are lost at handover and at night, there are many patients for one nurse, and a physician may not be available at once. In the parts of the Standard and of the sepsis protocol that I reviewed, there is no early-warning score and no automatic early-warning tool. Recognition depends on a person looking at the right numbers at the right time.

4.4 Where the system fits

The system cannot replace the monitor, the nurse or the protocol, and it should not try to. Its place is narrow. It looks at all patients every hour in the same way, notices a combination of changes that is easy to miss at night or at handover, and raises a flag early enough that the steps of the protocol can start sooner: a repeated measurement, lactate and blood cultures, the physician's assessment. Everything after the flag stays with the clinicians and the national protocol.

Three consequences for the design follow from this. The first recipient of a warning is the nurse, because the nurse is the first link of the existing chain; the physician is reached through the nurse, as today. The warning must state its reasons and the age of the data, because the nurse has to decide quickly whether it is real. And the only advice the system gives is the first step that the nurse would take anyway: re-check the measurements and inform the responsible physician. Chapter 7 shows how this is built.

5 Technology and method

This chapter justifies the main technical decisions. Each was taken by comparing alternatives against the requirements and the constraints of the project: one developer, one laptop, open data, and a system that must be tested and demonstrated end to end. The full comparison with the criteria is in the repository.

5.1 Architecture and component interaction

The system is built as a modular monolith: one backend process with a core package for the data and model logic, a REST API, and a dashboard that is a client of that API. A set of services connected by a message queue was considered and rejected. With one backend and one client there is nothing to scale independently, and every extra service would add deployment and testing work without serving a requirement. The interaction uses REST over HTTP with JSON, described by OpenAPI. gRPC would give stricter contracts and better performance, but the dashboard and the tests only need simple request–response calls, and JSON is easy to read in the API examples and the logs. A message queue would suit a push model with many consumers, which is not the case here.

The decision that matters most for the target setting is the data-source interface. The core asks one question, "which new values are there for the patients in the unit up to hour *t*", and two adapters answer it. In the diploma, a replay adapter reads the PhysioNet files and releases, at simulated hour *t*, only the rows recorded up to that hour. In a hospital, an MIS adapter would receive HL7 v2 messages or call the MIS interface. Everything behind the interface is the same code in both cases.

5.2 Backend, storage, dashboard and execution

The backend uses FastAPI rather than Flask or Django REST Framework. FastAPI validates requests from typed models and generates the OpenAPI description automatically, which serves the requirements on input validation and model metadata directly, and the machine-learning code runs in the same Python process. Flask would need extensions for the same result, and Django brings an ORM, templates and an admin interface that the project does not need.

Application data are stored in SQLite through SQLAlchemy, and the dataset and the models stay files. SQLite needs no server and is enough for one user on one laptop. It is not enough for a hospital, where several workstations would write acknowledgements at the same time, so the target system uses PostgreSQL. Because the application talks to the database only through the ORM, this change does not touch the application code.

The dashboard is built with Streamlit and Plotly. A React application would give more control over the interface, but it would also add a second language and a build pipeline for a single developer. Streamlit keeps the dashboard in Python, and because it only calls the API, no analytical logic lives in it and everything that matters can be tested through the API. The system runs in a virtual environment with a Makefile for the common commands; Docker Compose is optional. This keeps the setup simple and the demonstration free of cloud costs. Python 3.10–3.12 is required, because the pinned versions of NumPy and SHAP do not install on newer versions.

5.3 Machine-learning method

Data. The project uses the public training data of the PhysioNet/CinC Challenge 2019 [12]: 40,336 ICU stays from two hospital systems, recorded hourly, under the CC BY 4.0 licence. The

data are open and de-identified, have a defined label, and come from two systems, which allows an external test.

Task and label. For every patient-hour the model estimates the probability that `SepsisLabel` equals 1. The data owners set this label six hours before the onset of sepsis by the Sepsis-3 criteria [11], so the prediction horizon is fixed by the dataset and comparable with published work. The 426 stays that are positive from the first hour cannot be warned about in advance and are reported separately.

Split. Set A is split by patient into 70% for training, 15% for validation and 15% for the internal test, stratified by sepsis with a fixed seed; the lists of patient identifiers are stored. Set B is used only as the external test. No patient appears in two parts, and set B is never used for any decision, including the choice of excluded variables.

Leakage. Temporal leakage, using information that would not be available at the time of the prediction, is the most common way such results are inflated. Features for hour t use only hours up to t; missing values are filled forward within a patient only; any preprocessing is fitted on the training data. A unit test checks the first rule directly: appending future hours to a patient must not change the features of any earlier hour.

Features. Every retained variable enters with its last observed value, filled forward within the patient. Missingness is represented by a flag for whether the variable was measured in this hour, the hours since its last measurement and, for laboratory values, the number of measurements so far; a test that is ordered more often may itself be a sign of concern. For HR, O2Sat, Temp, SBP, MAP and Resp the model also receives the mean, minimum, maximum and standard deviation over the last six and twelve hours and the change against six hours earlier, because deterioration shows as a trend rather than a single value. Age, sex, the time from hospital to ICU admission and the hour since ICU admission give the context. EtCO2, TroponinI and Bilirubin_direct are excluded because they are never measured in more than 90% of set-A stays, and the unit identifiers Unit1 and Unit2 because they are missing in 46.8% of set-A stays; this was decided on set A only. Septic stays tend to be longer, so the hour since ICU admission may carry a signal specific to this dataset; a LightGBM variant without it is reported as a check.

Models. The baselines are a partial SIRS rule, computed from temperature, heart rate, respiratory rate, PaCO2 and the white cell count, with an alert at a score of two or more, and an L2-regularised logistic regression on the same features. qSOFA and NEWS2 cannot be computed, because the data contain no mental status. The main model is LightGBM, a gradient-boosted tree model that handles missing values natively, trains quickly on a laptop and can be explained per prediction with SHAP values. A recurrent model (GRU) is an optional extension after the main evaluation.

Metrics and success rule. The evaluation reports AUROC and AUPRC (with the prevalence as a reference), sensitivity, specificity and PPV at the operating point, calibration, and, per patient, the share of septic stays alerted before onset, the median lead time and the number of alerts per patient-day. The operating point is the highest threshold that still gives a sensitivity of at least 0.80 on the validation data, which makes the results comparable with Moor et al. [16]. Confidence intervals come from 1,000 bootstrap resamples of patients. The primary criterion is that LightGBM has a higher AUPRC than logistic regression on the internal test, with a 95% confidence interval of the paired difference that excludes zero. The drop on set B is reported for every model, whatever the result.

6 Requirements

The requirements were written after the literature review, the analysis of existing systems and the first user evidence, and revised after the supervisor meeting on 23 October. Each requirement has a pass/fail acceptance criterion, a priority and a source. The sources are the literature (LIT), assumptions of the evidence plan (AS), findings from the interviews (EV), properties of the dataset (DATA) and the guidelines or the supervisor (GUIDE). The full specification with all acceptance criteria is in the repository; this chapter summarises it and explains the choices that shaped it.

6.1 How priorities were set

Priorities follow the MoSCoW scheme. A requirement is Must if the system cannot answer its research questions or cannot be used safely without it: data validation, leakage-safe features, training and evaluation, the baselines, the prediction API, the alert rules, the patient list and trends, and the data-source interface. It is Should if it makes the system usable and trustworthy in the clinical setting but the research questions can be answered without it: explanations, the "insufficient data" state, stored predictions, the comparison view, model metadata, the fixed prompt and the acknowledgement. Could requirements are extensions that are started only after all Must requirements are done. Two features are explicitly excluded: a live connection to a hospital information system, which is designed but not built, and any treatment or medication advice.

6.2 Functional requirements

Table 6.1 lists the functional requirements in short form.

Table 6.1 – Functional requirements

ID	Requirement	Priority
FR-01	Import patient files in the Challenge format and validate their schema	Must
FR-02	Build features for hour *t* only from data recorded up to hour *t*	Must
FR-03	Train the baselines and the main model with one command from a configuration file	Must
FR-04	Evaluate every model on the internal and external test sets under one protocol	Must
FR-05	Compute the partial SIRS score from the last observed values	Must
FR-06	Return a risk estimate for a patient's history up to hour *t* through the API	Must
FR-07	Raise an alert at the threshold, suppress repeats for a window, raise a new alert when the risk rises by a margin	Must
FR-08	Show the patient list ordered by urgency, with the risk as a percentage and a coloured label	Must
FR-09	Show trends of five vital signs with the risk, threshold, alerts and the time since each measurement	Must
FR-10	Show the five measurements that contributed most to each risk	Should
FR-11	Report "insufficient data" instead of a risk when key vital signs are older than a limit (default 4 h)	Should
FR-12	Store each prediction with time, patient, risk, alert state and model version	Should
FR-13	Show a model comparison view with the evaluation results	Should
FR-14	Return metadata of the model in use through the API	Should
FR-15	Train a recurrent model (GRU) for comparison	Could
FR-16	Report the official Challenge utility score	Could
FR-17	Live integration with hospital information systems (designed, not built)	Won't

ID	Requirement	Priority
FR-18	Treatment or medication recommendations	Won't
FR-19	Show one fixed prompt with every alert	Should
FR-20	Acknowledge an alert with one of three outcomes; mark alerts not acknowledged in time	Should
FR-21	Take data through a data-source interface; replay the PhysioNet files hour by hour; run an hourly cycle	Must

Four acceptance criteria show how the requirements are made testable. For FR-02, appending ten future hours to a patient must leave the features of every earlier hour unchanged. For FR-07, with a six-hour window, a margin of 0.10 and a risk above the threshold at hours 10 (0.40), 12 (0.42) and 13 (0.55), alerts must be raised at hours 10 and 13 only, and without a further rise the next alert is possible from hour 19. For FR-11, a patient whose last key vital-sign measurement is more than four hours old must receive the state `insufficient_data`, no risk and no alert, and changing the limit to one hour in the configuration must change the result without code changes. For FR-21, running the hourly cycle for hours 1 to *t* must give the same features and predictions as building them directly from the rows up to *t*, and the replay adapter must never return a row from a later hour.

6.3 Non-functional requirements

Table 6.2 – Non-functional requirements

ID	Quality	Acceptance criterion	Priority
NFR-01	Reproducibility	The same configuration and seed give the same metrics (±0.001); the split is stored as lists of patient identifiers	Must
NFR-02	Data protection	No dataset files, archives or secrets in Git; no names or contacts in the evidence	Must
NFR-03	Performance	A prediction for the longest test stay is answered within 500 ms at the 95th percentile on the development laptop (to be confirmed)	Should
NFR-04	Maintainability	Test coverage of the core and the backend of at least 70% (to be confirmed)	Should
NFR-05	Portability	A fresh Python 3.10–3.12 environment and the README steps start the API and the dashboard	Must
NFR-06	Safe interpretation	Every response and screen with a risk shows "Research prototype — not a diagnosis"	Must
NFR-07	Usability	Risk, trends and reasons visible without scrolling at 1366×768; at least 2 of 3 respondents explain a high risk from the screen	Should
NFR-08	Transparency	A model card describes data, intended use, limitations and metrics	Should
NFR-09	Robustness	Malformed input returns a structured 4xx error without a stack trace	Should
NFR-10	Accountability	Views of a patient, acknowledgements and changes of settings create audit records that the API cannot change	Should

6.4 Changes after the meeting on 23 October

The meeting and the interviews changed the specification in two ways. The five changes proposed from the user evidence were adopted: FR-07 gained the re-alert margin, FR-08 the percentage with a coloured label and the order by urgency, FR-09 the time since each measurement, FR-11 the four-hour default, and FR-19 the fixed prompt. Three requirements were

added to describe how the system works in an ICU: FR-20 for the acknowledgement of alerts, FR-21 for the data-source interface and the hourly cycle, and NFR-10 for the audit trail that the national requirements expect from medical systems. The thirty-minute limit for acknowledgement and the 0.10 re-alert margin are first values, to be checked with ICU staff.

7 System design

This chapter presents the system twice: as it would work in a Kazakhstani hospital (the target system) and as it is built in the diploma. The two share everything except the source of the data and the infrastructure, and the chapter states for each element which of the two it belongs to.

7.1 Context and use cases

Figure 7.1 shows the context. The primary users are the ICU nurse, who sees the patient list and the alerts and acknowledges them, and the ICU physician, who assesses alerted patients under the national protocol and reviews the history of alerts. The secondary user is a researcher or ML engineer who imports data, trains and evaluates models and calls the API. In the diploma the data come from the PhysioNet files; in the target system they come from the hospital MIS and the bedside monitors. The use cases cover viewing the patient list and a patient's trends, seeing why the risk is high, seeing alerts and data-quality warnings, acknowledging an alert, reviewing alert history, and the researcher's tasks of importing, training, evaluating, configuring and calling the API.

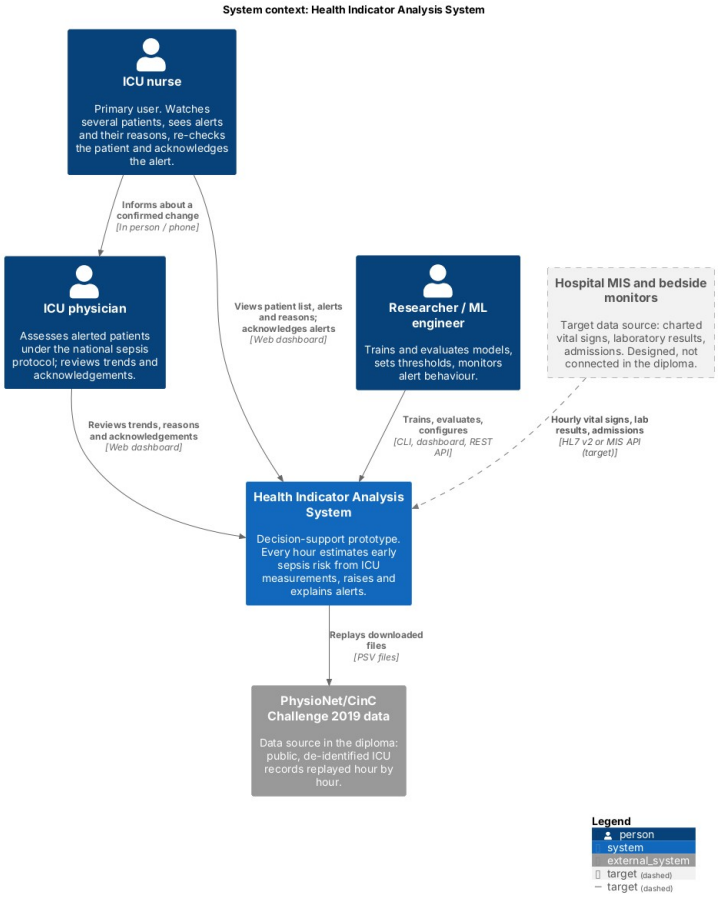


Figure 7.1 – System context

7.2 Where the data come from and how the indicators are measured

In the target setting the system reads three kinds of data from the MIS. The first are the vital signs charted by the nurse every hour. Charted values are preferred over raw monitor signals, because the nurse has already checked them and removed obvious artefacts such as a displaced saturation probe; the system works with the same numbers that the staff consider valid. The second are laboratory results, as soon as the laboratory releases them. The third are admission data: age, sex and the time of ICU admission. The data arrive through HL7 v2 messages, ADT for admissions

and transfers and ORU for observations and results, or through the MIS's own interface, depending on what the hospital provides. Both routes are allowed by the national requirements.

The PhysioNet data have the same shape: one row per patient-hour with the vital signs and laboratory values recorded in that hour and empty cells where nothing was measured. Each column corresponds to a real source. HR, O2Sat, Temp, SBP, MAP, DBP and Resp come from the bedside monitor through the nurse's chart, EtCO2 from capnography in ventilated patients, the 26 laboratory values from the laboratory, and Age, Gender, HospAdmTime and ICULOS from the admission record. This is why a replay of the files is a fair stand-in for a live feed: hour by hour, the system sees the same kind of record that a hospital would send.

The system does not measure anything itself. What it does is derive indicators from what was measured: the last value of each variable, how long ago it was measured, how it changed over the last six and twelve hours, and whether it is measured more often than before. These derived indicators are the features of the model (Section 5.3), and the same values are shown to the nurse next to the risk, so that the user and the model look at the same evidence.

The replay adapter has one rule: at simulated hour t it releases only the rows with ICULOS $\leq t$. The pipeline therefore cannot see the future, which is the same guarantee that the feature builder checks with its own test. The rule is tested directly (FR-21).

7.3 The hourly cycle

The system works in an hourly cycle, which matches the ICU recording schedule described by R2 and can be configured. For each patient in the unit the cycle performs eight steps (Figure 7.2):

1) Read: the adapter returns the new values since the last cycle.
2) Validate: values are checked against the schema and plausible ranges. A rejected value is logged with patient, hour and field and does not stop the cycle for other patients.
3) Check freshness: if none of the six key vital signs (HR, O2Sat, Temp, SBP, MAP, Resp) was recorded in the last four hours, the patient gets the state "insufficient data". No risk is computed and no alert can fire; the screen asks for a new measurement.
4) Build features from the hours up to t, with the same code that was used in training.
5) Compute the risk with the model in use, together with the five measurements that contributed most.
6) Apply the alert rules: an alert fires when the risk reaches the threshold; after any alert, a repeat for the same patient is suppressed for six hours; inside that window a new alert fires if the risk has risen by at least ten percentage points above the risk at the last alert, and it starts a new window.
7) Store the prediction with its time, model version, state and alert flag, and write an audit record.
8) Publish the new state: the dashboard reads it at its next refresh, every minute.

For a unit of six to twelve beds the cycle takes seconds; the time limit that matters is clinical, one hour, not technical. Failures are handled so that they cannot look like a stable unit. If the data source cannot be reached, the cycle records the failure and computes nothing new, and the dashboard shows the time of the last successful cycle. If the failure lasts, the patients pass into "insufficient data" after four hours, so a broken feed becomes visible to the staff instead of leaving old risks on the screen.

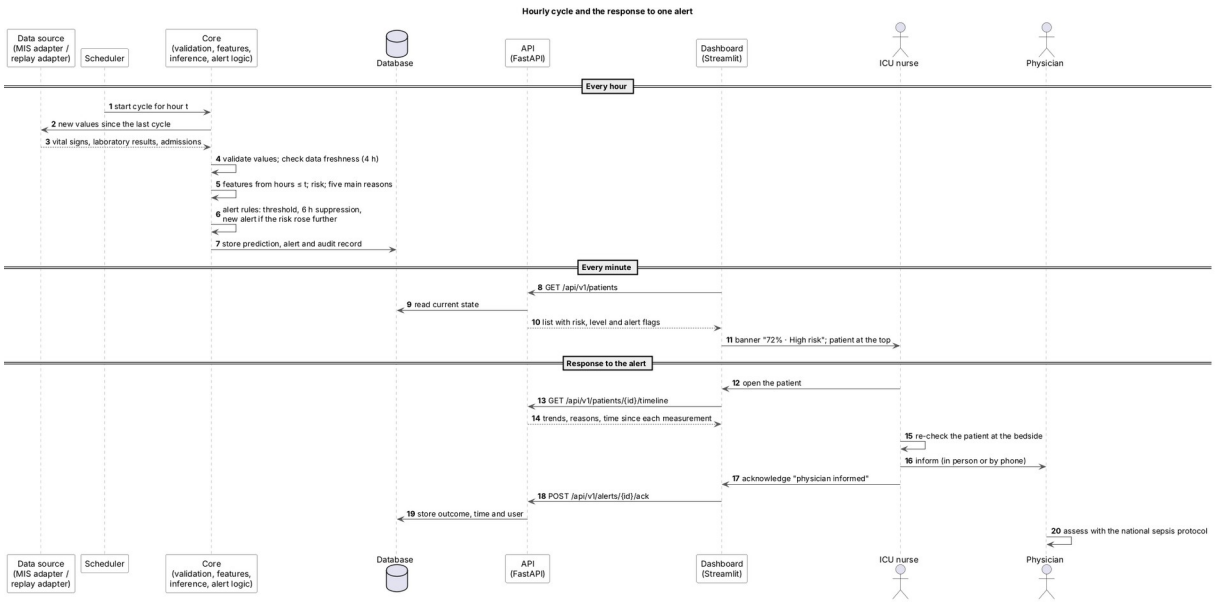


Figure 7.2 – Hourly cycle and the response to one alert (sequence diagram)

7.4 How the warning reaches the staff and what happens next

The dashboard runs in a browser on the ICU workstation at the nurse station. The patient list is ordered by urgency: unacknowledged alerts first, then acknowledged alerts, then the remaining patients by risk, and patients with insufficient data last. Each row shows the risk as a percentage together with a coloured label, for example "72% · High risk" in red, so that both readings asked for by the respondents are available. A new alert also shows a banner at the top of the screen until it is acknowledged. The patient view shows the risk curve and five vital signs on one time axis with the threshold and the alert hours, the last value of each vital sign with the time since it was measured, the five measurements that raised or lowered the risk in plain words, the fixed prompt "Re-check the measurements and inform the responsible physician", and the line "Research prototype — not a diagnosis".

Figure 7.3 shows the workflow with the system. When the alert appears, the nurse opens the patient and reads why the risk rose and how old each value is. The nurse then goes to the bedside, checks the patient and repeats the measurements; this is the same first step that R1 described for any suspected deterioration. The nurse acknowledges the alert with one of three outcomes: "re-checked, physician informed", "re-checked, change not confirmed" (for example a measurement error) or "already known and being treated". The acknowledgement clears the banner, and its time, user and outcome are stored. If the physician was informed, the physician assesses the patient under the national sepsis protocol, and the system takes no part in that decision. New measurements and laboratory results enter the MIS, and at the next cycle the risk is recomputed, so the physician sees whether the trend is turning. An acute emergency such as shock or cardiac arrest is handled by the hospital's existing emergency procedures, outside the system.

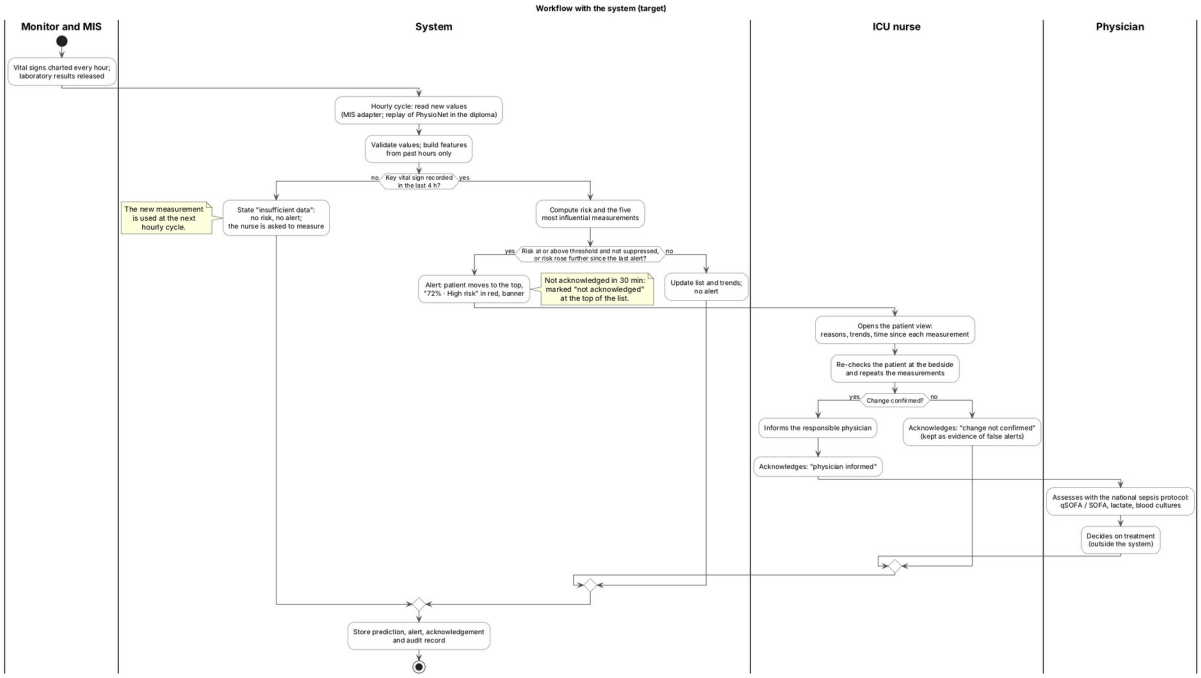


Figure 7.3 – Workflow in the ICU with the system (activity diagram)

If an alert is not acknowledged within thirty minutes, it stays at the top of the list marked "not acknowledged". In the target hospital the same event could also notify the duty physician through the MIS messaging; this is part of the target design only, and the diploma shows the marker on the dashboard. The acknowledgement outcomes are also evidence: the share of alerts marked "change not confirmed" is a direct, local measure of false alerts that complements the retrospective PPV from the evaluation.

7.5 Components

Figure 7.4 shows the components in four layers. The dashboard has three views: the patient list, the patient detail and the model comparison. The API has routers for patients, predictions, alerts and models; it validates every request and returns structured errors. The core package contains the data-source interface with its adapters, data validation, the feature builder, the inference service, the alert logic and the explanation component; the scheduler starts the hourly cycle and marks alerts not acknowledged in time. An offline pipeline for ingestion, training and evaluation uses the same core, so a feature is computed by the same code in training and in use. The storage layer holds the application database, the model store with metadata, the dataset files outside Git and the evaluation reports.

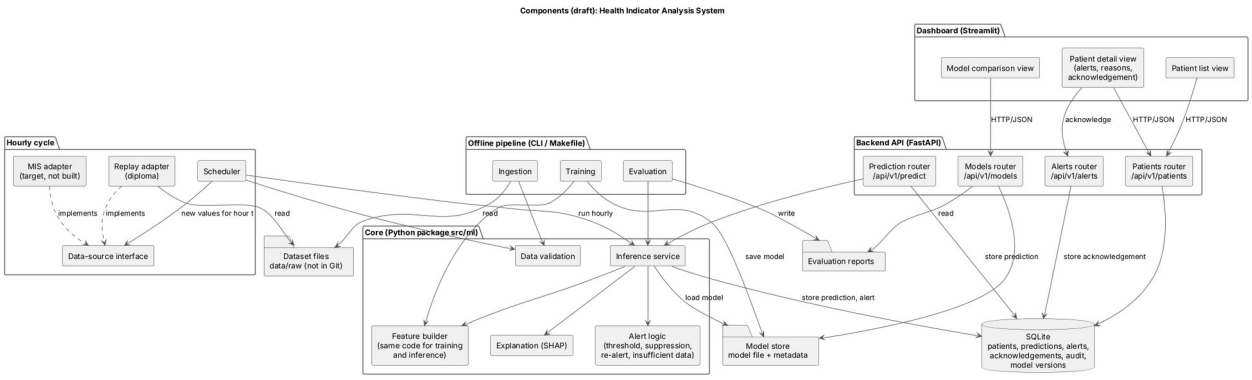


Figure 7.4 – Components of the system

Two dependency rules keep the structure testable. The dashboard calls only the API and contains no analytical logic, so all behaviour that matters can be tested through the API. The core depends on the data-source interface, not on a concrete adapter, so the replay adapter and a future MIS adapter can be exchanged without changes elsewhere.

7.6 Data model

The data model has eight entities (Figure 7.5). A Patient has one Measurement row per ICU hour, in the same shape as the PhysioNet files. Every hourly cycle writes a Prediction for each patient with the state (ok or insufficient data), the risk and its level, the five main reasons and the ModelVersion that produced it. A prediction that passes the alert rules raises an Alert, which keeps its reason (threshold or a further rise), the hour until which repeats are suppressed, and its status. The nurse closes it with an Acknowledgement that stores the outcome, the User and the time. Every view of a patient, every acknowledgement and every change of a setting leaves an AuditRecord. In the diploma these tables live in SQLite and in the target hospital in PostgreSQL. Demo patients are taken from the test sets only, so the dashboard never shows a patient the model was trained on.

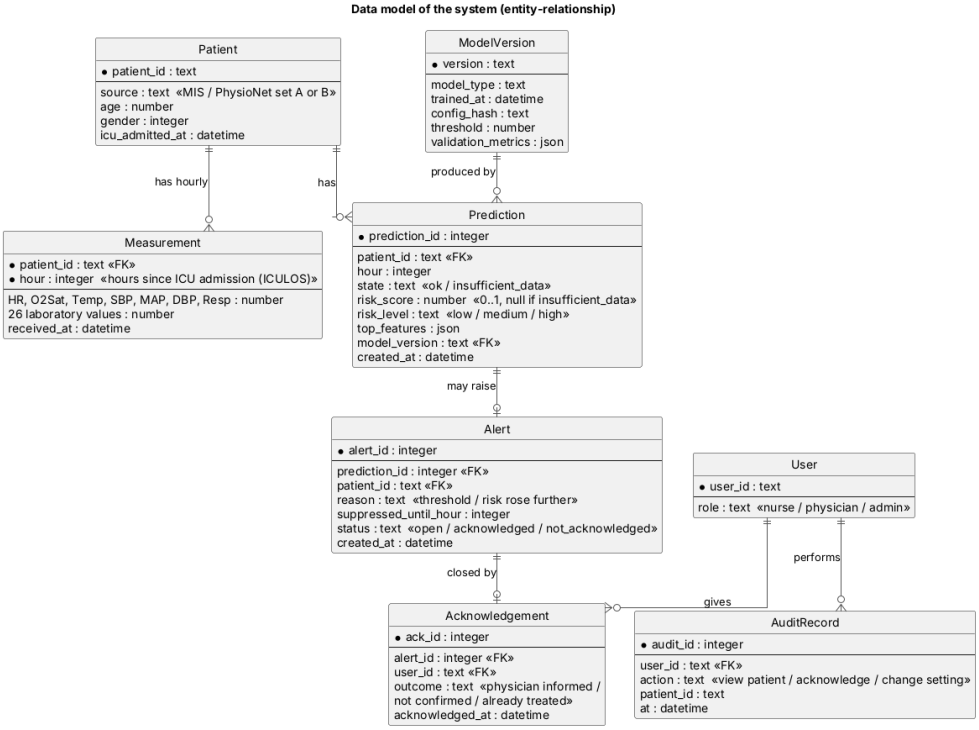


Figure 7.5 – Data model (entity–relationship diagram)

7.7 Deployment, security and introduction

Figure 7.6 shows both deployments. In the target hospital everything runs inside the hospital network, on an application server in Kazakhstan, as the national requirements demand. The MIS adapter receives HL7 messages or polls the MIS interface, a scheduler starts the hourly cycle, the FastAPI backend contains the core and the model, PostgreSQL stores the data, and the Streamlit server delivers the dashboard over HTTPS to browsers at the nurse station and in the physicians' room. Users log in with roles: nurses see the list and acknowledge alerts, physicians also see the history of acknowledgements, and an administrator sets thresholds and manages model versions. In the diploma the same components run on one laptop, with the replay adapter in place of the MIS adapter, SQLite in place of PostgreSQL and a researcher in the role of the administrator.

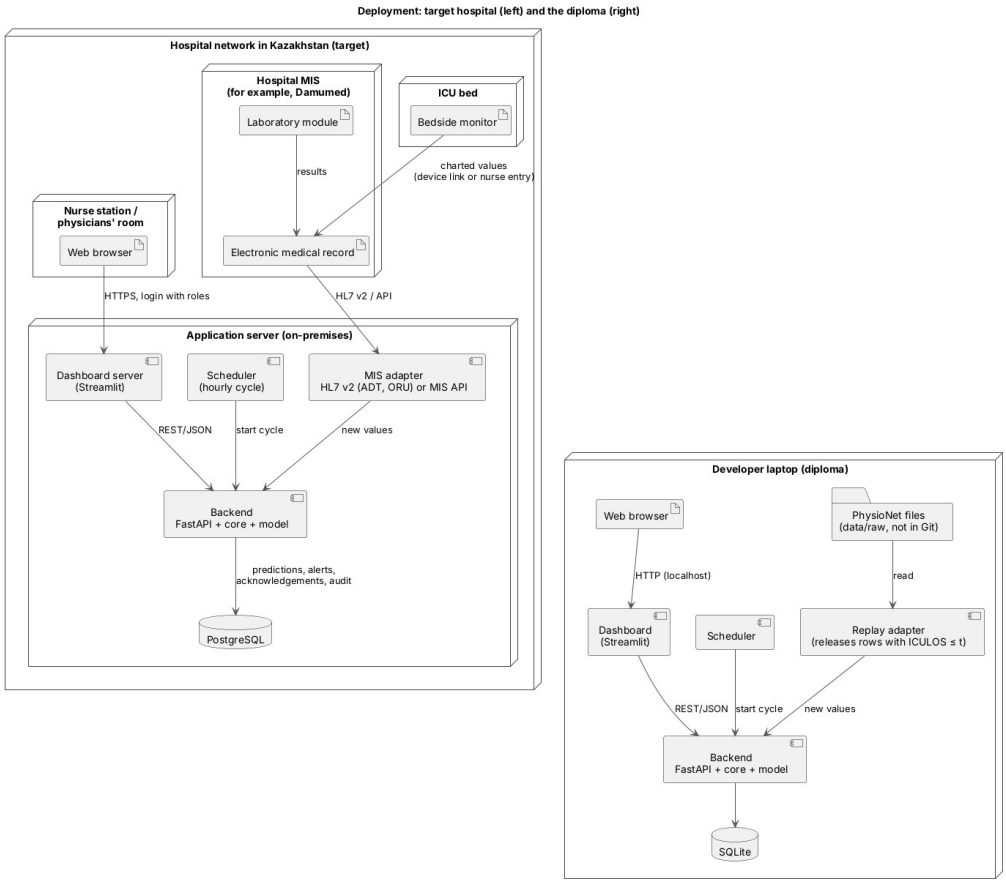


Figure 7.6 – Deployment in the target hospital and in the diploma

A hospital would introduce the system in three steps, as recommended by the implementation literature [20] and as done for CHARTwatch [21]. In silent mode, predictions are computed and stored but not shown for several weeks, which checks data completeness, alert frequency and agreement with real events. A pilot on one unit follows, with training of the staff, which R1 named as a condition for use. A review of alert frequency, acknowledgement times, the share of unconfirmed alerts, uptime and model drift, first weekly and then monthly, comes before any wider use.

The diploma implements the data-source interface with the replay adapter, the hourly cycle, validation, features, models and explanations, all alert rules, the acknowledgement with the "not acknowledged" marker, the API, the dashboard, and storage with audit records. It designs but does not implement the MIS adapter, notifications to phones or through the MIS, hospital accounts with single sign-on, and the silent mode and pilot, because they need a real hospital. A hospital would add one adapter rather than change the system.

7.8 Design decisions and rejected alternatives

Several decisions in this chapter were taken against reasonable alternatives, and the reasons are part of the design.

Hourly batch rather than streaming. The model could be run on every new value as it arrives. The ICU records vital signs every hour, the label of the dataset is defined per hour, and the respondents described no need for faster updates, so an hourly cycle matches the data and the work. A streaming design would add infrastructure without a requirement behind it.

Passive list and banner rather than pop-ups or pagers. A pop-up interrupts whatever the nurse is doing and is easy to dismiss without reading; a pager message needs devices and an

integration that the diploma cannot test. Following TREWS and CHARTwatch, the alert is shown where the nurse already looks, the patient list, and stays visible until it is acknowledged. A phone notification is kept as an option for the target hospital.

Suppression with a re-alert rather than a fixed window. A fixed window, as in CHARTwatch, would hide a patient whose risk keeps rising; no window would repeat the same alert every hour. The six-hour window with a re-alert after a rise of ten points follows both respondents, who asked for repeats only when the state changes.

Acknowledgement with an outcome rather than a dismiss button. A dismiss button would clear the screen but leave nothing to learn from. Three fixed outcomes take one click and make false alerts and response times measurable, which is what the introduction plan needs.

Stored predictions rather than predictions on demand. The dashboard reads the predictions written by the hourly cycle instead of asking the model each time a page opens. Every user then sees the same risk for the same hour, every alert can be traced to a stored prediction, and opening a page cannot fail because of the model. The prediction endpoint remains for tests and external callers.

8 Prototype

The prototype is a clickable dashboard built with the same technology as the planned system, Streamlit and Plotly. It runs on synthetic data: generated vital signs for eight demo patients and a hand-written risk formula stand in for real patients and the trained model, and every screen says so. It applies the same alert and data rules as the real system, and these rules are covered by automated tests. Its purpose is to show how a user works with the system and to collect feedback before the real pipeline exists. It is started with `streamlit run src/frontend/prototype_app.py`.

8.1 Patient list

The list answers the first question of a shift: whom to check first (Figure 8.1). Alerts that nobody has acknowledged come first, then acknowledged alerts, then the other patients by risk. The risk is shown as a percentage with a coloured label. A patient whose vital signs are older than four hours shows "Insufficient data" instead of a number and is listed last. The caption states the alert rules in use.

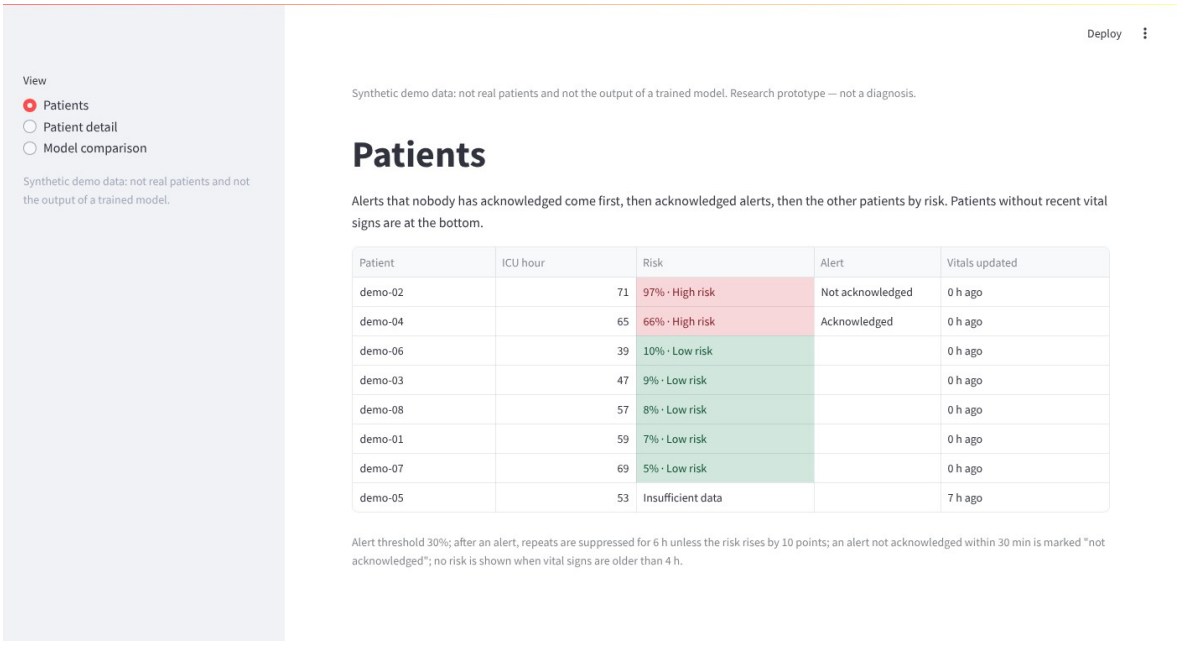


Figure 8.1 – Patient list ordered by urgency

8.2 Patient view with an alert

Figure 8.2 shows a patient at the hour of the first alert. The banner states why the alert fired and gives the fixed prompt. Below it the nurse chooses one of the three outcomes and acknowledges the alert; an alert from an earlier hour that nobody acknowledged is marked as such. The row of last values shows how old each measurement is. The chart shows the risk and five vital signs on one time axis, with the threshold and triangles at the alert hours, and a slider replays the stay hour by hour. The right panel shows which measurements raised or lowered the risk and reminds the user what the model does not see: notes, medications, examinations and anything not measured hourly.

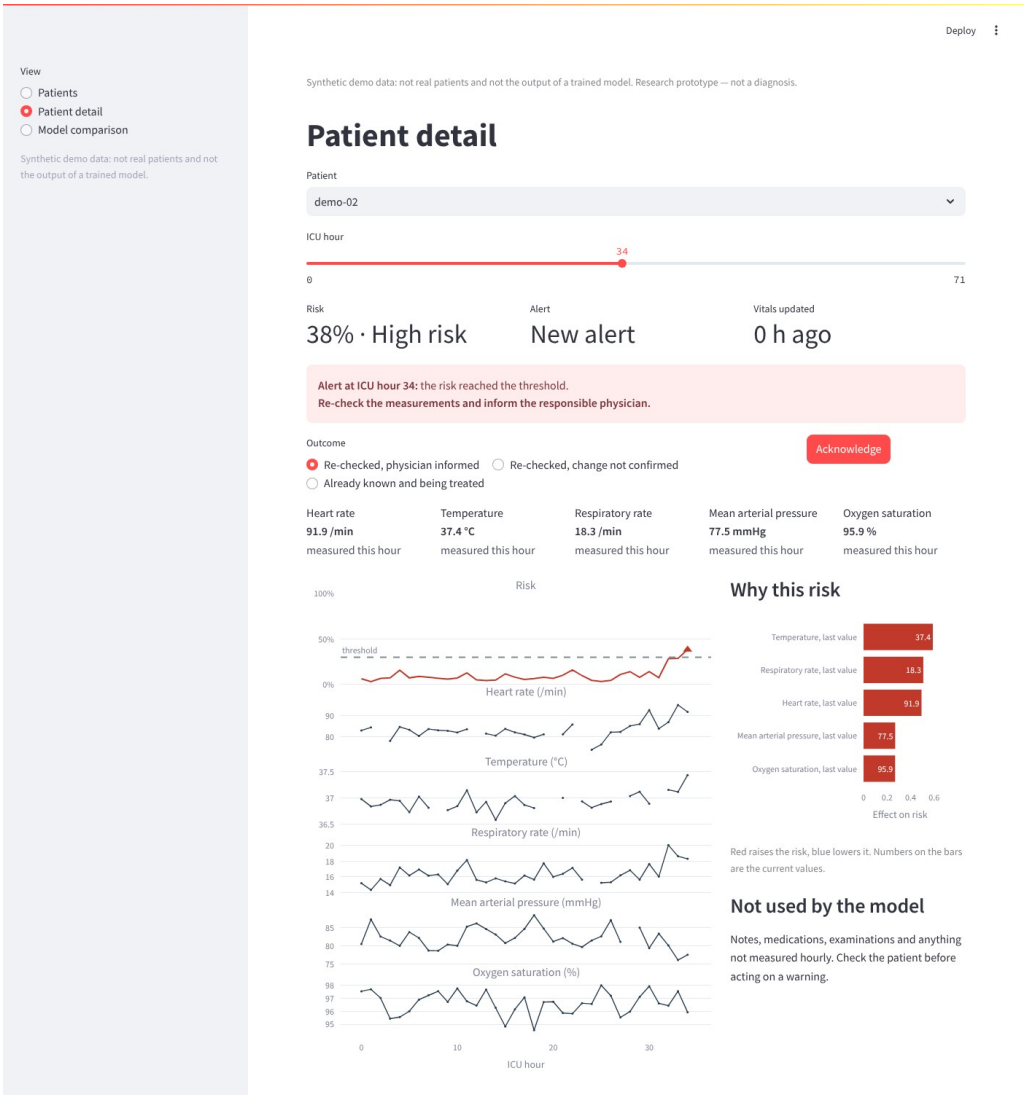


Figure 8.2 – Patient view at the first alert, with the acknowledgement

When the key vital signs have not been measured for more than four hours, no risk is shown, no alert can fire, and the screen asks for a new measurement (Figure 8.3). This follows the "indeterminate" output of COMPOSER: it is better to say that the data are not sufficient than to show a confident number built on stale data.

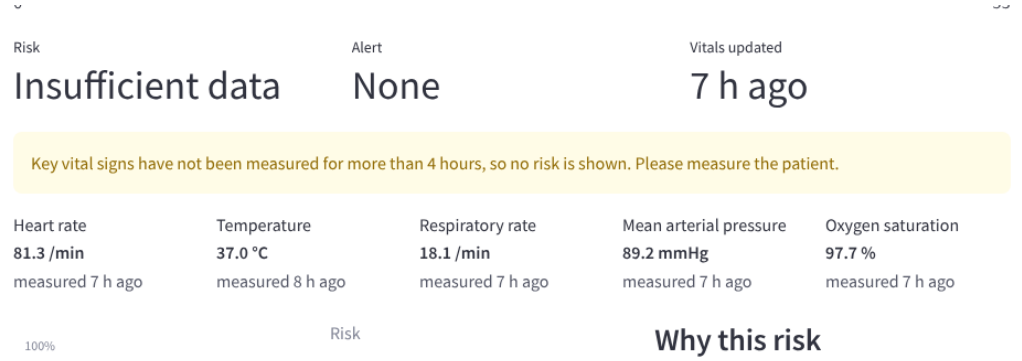


Figure 8.3 – Patient with insufficient data (upper part of the patient view)

8.3 API contract

The dashboard will read everything from the backend through the REST API in Table 8.1. Every response that contains a risk also contains the disclaimer, invalid requests return status 422

with errors for each field, and acknowledging an alert twice returns status 409. Request and response examples are in the repository.

Table 8.1 – Endpoints of the API

Endpoint	Purpose	Requirements
`GET /api/v1/health`	Service status	NFR-05
`POST /api/v1/predict`	Risk, label, alert, five reasons and data state for a history up to hour *t*	FR-06, FR-07, FR-10, FR-11
`GET /api/v1/patients`	Patients in order of urgency with risk and alert state	FR-08
`GET /api/v1/patients/{id}/timeline`	Hourly measurements and risk of one patient	FR-09
`GET /api/v1/patients/{id}/predictions`	Stored predictions with model version	FR-12
`GET /api/v1/patients/{id}/alerts`	Alerts of one patient with their acknowledgements	FR-20
`POST /api/v1/alerts/{id}/ack`	Acknowledge an alert with an outcome	FR-20, NFR-10
`GET /api/v1/models/current`	Type, version, threshold and validation metrics of the model	FR-14

8.4 Known gaps

The prototype does not yet meet the usability criterion of NFR-07. At 1366×768 the patient view is about 1,680 pixels high, so the risk, the alert banner and the last values are visible without scrolling but the trends are not; the real dashboard will use a more compact layout. In the prototype the acknowledgement is kept only in the browser session, whereas the real system stores it with the user and the time. The screens have not yet been shown to respondents: the survey of 21 October was sent before they were published, so the user check is planned with the next respondents.

9 Verification and evaluation plan

The aim of the project is demonstrated in three ways. Tests show that the software does what the requirements say. The model is evaluated against two baselines on an internal and an external test set. And users check whether the screens explain a warning and whether the response path fits their work. This chapter describes each of them.

9.1 Testing the software

Unit tests check single functions with small patient files written inside the tests: schema validation, the feature builder, the partial SIRS score, the alert rules, the data-quality rule and the replay adapter. Integration tests run the pipeline from raw files to features and predictions on a fixed sample of 50 patients from the training split, selected by a stored list of identifiers; the data stay in the ignored data folder, and the tests skip with a clear message when the data are not present. API tests use FastAPI's test client to check the endpoints, the validation of requests, the error format and the presence of the disclaimer. No test fixture in the repository contains rows copied from the dataset.

Above these sit acceptance tests, one for each Must and Should requirement, written from the point of view of the user or the researcher, and inspections for what a test cannot fully decide: the content of the repository, the documentation, and the disclaimer on every screen. Appendix B traces every Must and Should requirement to its use case, component and test. Three examples show the idea. The test for FR-02 builds features for a stay, appends ten more hours and checks that the features of all earlier hours are identical. The test for FR-07 feeds the alert logic the risk values of the acceptance criterion and expects alerts at hours 10 and 13 only, the second with the reason "risk rose". The test for FR-21 runs the hourly cycle for 50 test patients and compares the result with predictions built directly from the rows up to each hour.

Two groups of tests run today: six tests for the data-analysis script, covering schema rejection, the derived onset, measurement gaps and duplicate identifiers, and six tests for the alert and data rules of the prototype, which encode the acceptance criteria of FR-07, FR-08, FR-09 and FR-11 and will be reused for the real alert module. A failing check becomes a YouTrack task linked to its requirement and is closed only after the check passes. A pre-defense is ready when all Must acceptance tests pass or have a documented and accepted reason, the inspections of data protection and of the disclaimer pass, and the setup from the README has been demonstrated in a clean environment.

9.2 Evaluating the model

The model is evaluated on two test sets. The internal test, 15% of the patients of set A that are never used for training or tuning, answers how well the model works on new patients from the same hospital system. The external test, all of set B, answers how much performance drops at a different hospital system. All models are evaluated on exactly the same patients, features and rules.

At the level of hours the evaluation reports AUROC and AUPRC, with the prevalence of positive hours as the reference for AUPRC; sensitivity, specificity, PPV and F1 at the operating point; and the Brier score with a calibration curve in ten bins. At the level of patients it reports the share of septic stays with at least one alert before onset, the median lead time between the first alert and onset, the share of non-septic stays with an alert, and the number of alerts per patient-day. The last two are the alert burden that the staff would experience. Confidence intervals come from 1,000 bootstrap resamples of patients.

The baselines are the partial SIRS rule with a score of two or more and the regularised logistic regression on the same features. The success rule is fixed in advance: LightGBM must reach a higher AUPRC than logistic regression on the internal test, with a 95% confidence interval of the paired bootstrap difference that excludes zero. The comparison with partial SIRS, the drop from the internal to the external test for every model, and the patient-level alert metrics are reported whatever the result. A primary criterion that is not met is reported as a result; it does not hide the comparison.

The measurement conditions are recorded with every result: dataset version 1.0.0, the stored patient-level split with seed 42, the configuration file and its hash, the pinned library versions, and the hardware, a laptop without a GPU.

9.3 Evaluating the workflow

Whether users understand the warning is checked in a user check with two or three respondents, first on the prototype screens and later on the real dashboard. Each respondent is asked why the risk is high for the patient on the screen, what they would do next, and what is missing or confusing. Two further questions test the open values of the design: whether thirty minutes is a realistic time to acknowledge an alert, and whether a new alert after a rise of ten percentage points would come too often or too rarely. The check succeeds if at least two of three respondents explain the reason for the high risk correctly from the screen (NFR-07). The answers are recorded as transcripts and feed back into the requirements.

In a hospital the workflow would be evaluated during the silent mode and the pilot (Section 7.7) with measures that the design already records: the number of alerts per day, the time to acknowledgement, the share of each acknowledgement outcome, and the uptime of the hourly cycle. These measures are not part of the diploma's results, because the diploma does not run in a hospital, but the data model and the API are built to produce them.

10 Project management and risks

10.1 How the work is organised

Every piece of work is a task in the YouTrack project DIP with an assignee, a priority, a due date, an expected outcome and acceptance criteria. Each task gets its own branch from `main`, named `DIP-<n>-<name>`; commit messages start with `DIP-<n>:`; changes reach `main` only through a pull request, and branch protection on `main` enforces this. A task is closed when its acceptance criteria are met and the link to the pull request is added to it. Submissions are marked with the tags `pre-defense-1`, `pre-defense-2` and `final-pre-defense`, which are never moved. The project is individual, so I am responsible for all of the work; the supervisor reviews the key decisions at meetings, and each review is recorded in the feedback log.

Before 19 October 2026 the process was not followed consistently: four pull requests were merged without review, three of them into feature branches instead of `main`, and some commits reached `main` directly. This is recorded in the feedback log. The history has not been rewritten, and the rules above apply from 21 October.

10.2 Milestones

The dates after pre-defense 1 in Table 10.1 are my proposal and will be aligned with the official dates of pre-defense 2, the final pre-defense and the defense once they are announced.

Table 10.1 – Milestones

Milestone	Target date	Done when
M0 Pre-defense 1	24 Oct 2026	All eleven components merged; tag pushed; form submitted
M1 Data pipeline and baselines	20 Nov 2026	Validation, leakage test, data-source interface with replay, hourly cycle and both baselines pass their tests
M2 Main model and evaluation	18 Dec 2026	Internal and external results with confidence intervals; model card
M3 API and dashboard	26 Feb 2027	API and dashboard acceptance tests pass, including acknowledgement and audit; user check done
M4 Testing and documentation	26 Mar 2027	All Must tests pass; clean-setup demonstration; diploma text drafted

10.3 Main risks

The register in the repository lists fourteen risks with probability, impact, trigger and response. Six of them shape the plan most.

The user evidence may stay thin. Only two respondents have answered so far, below the target of three to five, and the evidence plan cannot force more answers. The response is to continue the survey with follow-up questions, to use the data analysis as additional evidence, and to report the actual number honestly.

Temporal leakage could make the results look better than they are. It is the most common error in this kind of work, and it is controlled by the patient-level split, forward filling within a patient only, preprocessing fitted on the training data, the dedicated unit test and, since 23 October, the replay adapter, which cannot release a future row.

The main model may not beat logistic regression, and performance on set B is likely to drop. Both outcomes are reported as results. The engineering contribution and the comparison remain valid in either case, and no claim is made beyond the evaluated data.

The scope may grow beyond what one student can finish. The MoSCoW priorities, the rule that Could requirements start only after all Must requirements are done, and the decision to design rather than build the hospital integration keep it bounded.

Two risks were added after the meeting on 23 October. The workflow of the target system may not match real ICU practice in Kazakhstan; it is based on the national documents and the interviews, its open values are marked as first values, and it is checked in the next discussions and, in a hospital, in silent mode first. And there may be no access to a hospital information system for an integration; the data-source interface keeps the system independent of the source, the diploma uses the replay adapter, and the MIS adapter is described as designed, not as built.

11 Supervisor feedback and how it was addressed

Every comment from the supervisor is recorded in the feedback log with the date, the channel, my decision and the change it led to, so that the path from review to revision can be followed in Git and YouTrack. This chapter summarises the three rounds of feedback before pre-defense 1.

11.1 Review of 19 October

On 12 October the supervisor asked me to send the materials that were ready, and his written feedback came on 19 October. He wrote that the project had a starting structure but that the documentation, particularly the README, needed substantial improvement. He asked me to explain what problem the system addresses, who its users are and why it is needed; to present my own vision of how the system should work; to justify the algorithms, the data and the evaluation; to compare the research and the existing systems critically; and to complete the required documents according to the pre-defense guidelines. He added that he wanted to hear my own reasoning for the technical decisions rather than only review formal documents, and that messages to him should begin with a greeting.

In response the README was rewritten, the proposal was rewritten with defined primary and secondary users, a section with my proposed solution was added together with the requirements, the architecture and the prototype, and the machine-learning methodology was written down with a reason for every decision. The literature review gained a table of limitations and responses, the inconsistencies between the system comparison and the literature matrix were corrected, and the components that were empty on 19 October were completed. My own summary of the reasoning behind the main decisions is still open; I am writing it in my own words.

11.2 Questions of 21 October

On 21 October I asked whether early sepsis prediction from ICU time-series data was an acceptable concrete focus for the agreed topic. After a clarification the supervisor replied: "Please prepare as research justification and we will see [it on] 23", and "Good idea". The research justification was prepared as a separate document with the burden of the problem, the evidence from the literature, the research gap, the research questions and hypotheses, and the feasibility. I also asked whether a survey together with an analysis of the dataset was acceptable evidence; this was discussed at the meeting.

11.3 Meeting of 23 October

At the Teams meeting on 23 October the supervisor reviewed report version 1. He said that the sepsis focus is a good idea, but that every aspect has to be thought through and planned in depth so that the work reads as a diploma project. His comments, and what was done about them, were the following.

He asked me to put the emphasis on the system as it should be: how it is logically built and how the whole process will run, so that the committee sees a deep understanding of the topic, how the system would be organised in reality, and my own interest in it. In response the target system was described separately from the diploma implementation and connected to it through the data-source interface (Chapter 7), with new diagrams of the clinical workflow, the alert sequence, the deployment and the data model.

He asked me to describe how the ICU process runs in Kazakhstan and what happens after a nurse or another user receives a notification. Chapter 4 now describes the organisation of intensive

care, the measurements and records, and the escalation under the national sepsis protocol, and Section 7.4 describes the response to an alert step by step, including the acknowledgement that was added as FR-20.

He asked how exactly the system shows alerts and data to nurses and other users, how it measures the indicators and where it takes the data from. Sections 7.2–7.4 answer these questions, the five changes from the user evidence were adopted in the requirements, and the prototype was updated to show them.

He asked me to add, for the existing systems, their ERD diagrams, their architecture or anything public that shows how they are built. No public ERD exists for any of the five systems, so Chapter 3 gives reconstructed architectures with explicit statements of what the sources do not describe.

He asked me to record the respondents' answers as transcripts. Both respondents agreed to the publication of their anonymised answers, which are given in full in Appendix A.

Finally, he asked me to rewrite the report so that it is more argued and deeper, with fewer tables and simpler, more academic formatting, and he said that the diploma must show either a very well-developed system from a software-engineering point of view or a very deep machine-learning methodology. I chose the software-engineering direction, and this report is the rewritten version: continuous text, tables only where values are compared, and the depth placed in the design of the system and its fit with the clinical workflow.

Conclusion

This report has presented the analysis, design and plan of a system that estimates the risk of sepsis for ICU patients from their hourly measurements and delivers each warning to the nurse with its reasons, a fixed first step and a recorded response. The evidence from two interviews and from the dataset shows that early recognition of deterioration is a real difficulty, that it depends on a nurse noticing a combination of changes, and that the data needed to support this are recorded every hour but are irregular and incomplete. The literature and the existing systems show that a model is only as useful as the workflow around it and that its performance has to be checked on new data and against simple scores.

The main result of this stage is a design that can be followed from the hospital's data to the nurse's response. The target system reads charted values and laboratory results from the hospital information system, runs an hourly cycle that validates the data, refuses to give a number on stale data, computes the risk with its reasons and applies alert rules derived from the respondents' answers, and shows the result at the nurse station, where every alert is acknowledged with an outcome and every action is logged. The diploma builds the same system on open data, replaced at one point, the data source, by a replay that cannot see the future. Nineteen functional requirements to be built, two that are explicitly excluded, and ten non-functional requirements connect the design to its verification; each requirement to be built has an acceptance test.

The work has clear limitations. The user evidence rests on two respondents, the workflow has not yet been checked with ICU nurses, and the prototype runs on synthetic data and does not yet meet the no-scrolling criterion. The pipeline, the model, the API and the connected dashboard are planned for the next milestones, and the system will not be connected to a hospital or used on patients.

The next steps are to continue the interviews and run the user check of the prototype, including the open values of the alert path; to build the data pipeline with the replay adapter, the hourly cycle and both baselines by 20 November; and then to train and evaluate the main model and build the API and the dashboard as planned.

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Appendix A. Interview transcripts

The transcripts below are the full answers of both respondents to the anonymous survey of 21 October 2026 and to the follow-up questions in a messenger. The Russian text is copied exactly, including typing errors and punctuation; the English translation is mine and follows the meaning. The respondents' names and the contact field of the survey are removed; nothing else is. Both respondents agreed to the publication of their anonymised answers in full. In the report the answers are cited as, for example, R1-Q7 (survey question 7 of respondent 1) or R2-FU4 (follow-up question 4 of respondent 2).

A.1 R1 — Nurse, general medical or surgical ward, more than 5 years

Survey, 21 Oct 2026, 17:03

Q4. When you need to know whether a patient is stable or getting worse, what do you look at first?

В первую очередь я смотрю на проходимость дыхательных путей,дыхание и конечно на кровообращение.И конечно же дальнейших изменений в состоянии пациента от исходного состояния.

Translation: First of all I look at airway patency, breathing and, of course, circulation. And of course at further changes in the patient's condition compared with the baseline.

Q5. Where does that information come from: monitors, observation charts, the hospital information system, lab results, something else?

В первую очередь я ориентируюсь на нынешнюю состоянию пациента.Сначала смотрим на мониторинг пациента ЧСС,артериальное давление и частоту дыхания,если есть возможность ЭКГ.После этого смотрим историю болезни пациента.Конечно результат анализов очень важен.Но в некоторых случаях это зависит от состояния пациента, если состояние резко ухудшается, помощь не откладывают ради просмотра документации или ожидания анализов.

Translation: First of all I go by the patient's current condition. First we look at the patient's monitoring: heart rate, blood pressure and respiratory rate, and an ECG if possible. After that we look at the patient's medical history. Of course test results are very important. But in some cases it depends on the patient's condition: if the condition deteriorates sharply, help is not delayed to read documentation or to wait for test results.

Q6. In practice, how is it usually noticed that a patient is getting worse? Who notices, and what happens next?

Чаще всего замечают медицинский персонал:медсестра,медбрат или же врач.Например медсестра,измеряя давление или же пульс замечает какие то изменение,то сразу же смотрит на пациента,проверяя в сознании ли пациент и проверяет жизненные показатели.При значимых изменениях сообщаем об этом врачу.А при критических ситуациях начинают необходимые манипуляции в пределах своей компетенции.

Translation: Most often it is noticed by medical staff: a nurse or a doctor. For example, when a nurse notices some change while measuring blood pressure or pulse, she immediately looks at the patient, checks whether the patient is conscious and checks the vital signs. If the changes are significant, we report them to the doctor. In critical situations the necessary procedures are started within one's competence.

Q7. Where does this process work poorly or fail? If you can, give an example without patient details.

Чаще всего сбои возникают на этапе выявления ухудшения состояния пациента, а также при передачи информации между сотрудниками.Например учащенное дыхание или же можем взять в пример необычную сонливость воспринимают как незначительные изменения.При смене дежурства иногда такие изменения могут остаться незамеченными.Иногда даже при ухудшение состояния пациента врачи могут быть недоступными.

Translation: Most often failures happen at the stage of detecting deterioration and when information is passed between staff. For example, fast breathing or unusual drowsiness are taken as minor changes. At a shift change such changes sometimes go unnoticed. Sometimes, even when the patient deteriorates, doctors may be unavailable.

Q8. Which of these are used where you work or had placements?

Automatic alerts in the hospital information system / Автоматические оповещения в медицинской системе

Q9. Do such scores and alerts help in practice? Why or why not?

Их эффективность зависит от точности измерений, своевременного внесения данных пациентов в систему и реакции медицинского персонала. Иногда возникают ложные тревоги или оповещения теряются из-за высокой нагрузки.

Translation: Their effectiveness depends on the accuracy of measurements, on timely entry of patient data into the system and on the response of the medical staff. Sometimes there are false alarms, or alerts get lost because of high workload.

Q10. Have you seen warnings or alerts being ignored? If so, why does that happen?

Я не могу утверждать, что лично наблюдала такую ситуацию, но в медицинской практике предупреждения могут игнорироваться.Ну во многом из за высокой нагрузки и нехватки медицинского персонала.Иногда даже из за частых ложных тревог,снижается внимание к оповещениям.

Translation: I cannot say that I have seen such a situation personally, but in medical practice warnings can be ignored. Largely because of high workload and staff shortages. Sometimes, because of frequent false alarms, attention to alerts goes down.

Q11. Imagine a system shows a risk-of-deterioration score for a patient. What would need to be shown next to that number for you to understand and trust it?

Я бы хотела видеть не просто цифру риска, а понимать,почему система её показывает. Например,какие показатели изменились,когда их измеряли и что именно вызывает опасение. Ещё важно, чтобы система подсказывала,на что нужно обратить внимание и какие действия предпринять. Тогда будет проще оценить состояние пациента и понять,насколько можно доверять этому предупреждению.

Translation: I would like to see not just a risk number but to understand why the system shows it. For example, which measurements changed, when they were measured and what exactly is of concern. It is also important that the system suggests what to pay attention to and what actions to take. Then it will be easier to assess the patient's condition and to understand how far this warning can be trusted.

Q12. Which of these would be useful to you?

Which measurements influenced the risk / Какие показатели повлияли на риск, A warning when data are missing or out of date / Предупреждение, если данных мало или они устарели, A list of patients sorted by risk / Список пациентов, отсортированный по риску

Q13. What would make you distrust such a system or avoid using it?

Я бы не доверяла системе, если бы она часто ошибалась,показывала непонятные результаты или не учитывала реальное состояние пациента. Также важно,чтобы данные обновлялись вовремя.Если система показывает высокий риск,а непонятно почему и что делать дальше,пользоваться ею было бы сложно. Для меня важно,чтобы она помогала в работе,а не усложняла её.

Translation: I would not trust the system if it often made mistakes, showed unclear results or did not take the patient's real condition into account. It is also important that the data are updated on time. If the system shows a high risk but it is unclear why and what to do next, it would be hard to use. For me it is important that it helps in the work and does not make it more complicated.

Is there anything important I did not ask?

Я думаю, важно учитывать загруженность медицинского персонала и то насколько легко пользоваться такой системой в повседневной работе.Она должна быть простой и понятной,быстро показывать важную информацию. Также необходимо обучить сотрудников работе с ней.Главное, чтобы система помогала вовремя заметить ухудшение состояния пациента и не заменяла опыт и профессиональное мнение медицинского персонала.

Translation: I think it is important to take into account the workload of medical staff and how easy such a system is to use in everyday work. It should be simple and clear and show important information quickly. Staff also need to be trained to work with it. The main thing is that the system helps to notice deterioration in time and does not replace the experience and professional judgement of medical staff.

Follow-up in a messenger: questions sent 21 Oct 2026, 19:59; answers received 22 Oct 2026, 06:28

1. Через сколько часов без новых измерений (пульс, давление, температура, дыхание, сатурация) в отделении Вы бы считали данные пациента устаревшими?

After how many hours without new measurements (pulse, blood pressure, temperature, breathing, saturation) on the ward would you consider a patient's data out of date?

Думаю если показатели не обновлялись 4–6 часов,система уже должна обратить на это внимание. Особенно важно учитывать состояние пациента,потому что в некоторых случаях даже за короткое время показатели могут сильно измениться.

Translation: I think that if the measurements have not been updated for 4–6 hours, the system should already draw attention to it. It is especially important to take the patient's condition into account, because in some cases the measurements can change a lot even in a short time.

2. Как часто можно повторять оповещение по одному и тому же пациенту, чтобы оно не начало раздражать?

How often can an alert be repeated for the same patient before it starts to annoy?

На мой взгляд не нужно постоянно отправлять одинаковые уведомления.Лучше повторять их,если проблема сохраняется или состояние пациента меняется.Но если ситуация опасная,система должна сразу сообщить об этом повторно,чтобы предупреждение не пропустили.

Translation: In my view, identical notifications should not be sent all the time. It is better to repeat them if the problem persists or the patient's condition changes. But if the situation is dangerous, the system should report it again at once, so that the warning is not missed.

3. Что понятнее при оценке риска: число (например, 0.7 или 70%), цвет (зелёный / жёлтый / красный) или слово (низкий / высокий)?

What is clearer for assessing risk: a number (for example 0.7 or 70%), a colour (green / yellow / red) or a word (low / high)?

Лично мне было бы понятнее если бы одновременно использовались цвет и короткая надпись например «Высокий риск» красным цветом. Так информацию можно быстрее понять,особенно когда много работы и нужно быстро ориентироваться.

Translation: Personally, it would be clearer for me if a colour and a short label were used together, for example "High risk" in red. That way the information is understood faster, especially when there is a lot of work and you need to orient yourself quickly.

4. Вы писали, что изменения могут остаться незамеченными при смене дежурства. Что чаще всего теряется при передаче смены и что помогло бы это не пропустить?

You wrote that changes may go unnoticed at a shift change. What is most often lost at handover, and what would help not to miss it?

Думаю при передаче смены можно упустить важные изменения которые произошли за ночь или информацию о том за какими пациентами нужно внимательнее следить.Было бы удобно если бы

система показывала последние показатели,их изменения и отдельный список пациентов требующих особого внимания.

Translation: I think that at handover one can miss important changes that happened during the night, or information about which patients need to be watched more closely. It would be convenient if the system showed the latest measurements, their changes and a separate list of patients who need special attention.

5. Вы писали, что система должна подсказывать, на что обратить внимание и что делать. Какие подсказки были бы полезны? Приведите пример.

You wrote that the system should suggest what to pay attention to and what to do. What prompts would be useful? Give an example.

Мне было бы полезно чтобы система не просто показывала уровень риска,а объясняла почему он повысился и на какие показатели нужно обратить внимание.Например если у пациента снизилась сатурация или изменились пульс и давление система могла бы выделить эти изменения и подсказать что необходимо проверить показатели повторно и сообщить ответственному медицинскому работнику.

Translation: It would be useful for me if the system did not just show the risk level but explained why it rose and which measurements need attention. For example, if a patient's saturation dropped or the pulse and blood pressure changed, the system could highlight these changes and prompt that the measurements need to be checked again and the responsible medical worker informed.

A.2 R2 — Resident or intern; ICU, urology–nephrology department, operating theatre; 1–3 years

Survey, 21 Oct 2026, 17:04

Q4. When you need to know whether a patient is stable or getting worse, what do you look at first?

Первую очередь жалобы пациента, объективное оценка состояние и анализы.

Реанимационные пациенты по состоянию гемодинамик, контроль и динамика анализов(КЩС, ОАК, БХА, ОАМ) это и дает эффективность лечение, инструментальные исследования ЭКГ, эхокг, КТ и Рентген. Если пациент на жизнеобеспечение (на аппарате ИВЛ, ЭКМО, ГДФ) изменения параметров и их не эффективность.

Translation: First of all the patient's complaints, an objective assessment of the condition, and test results. ICU patients — by the state of the haemodynamics, and by the monitoring and trends of tests (acid–base balance, complete blood count, blood biochemistry, urinalysis); this is what shows whether treatment works; and instrumental examinations: ECG, echocardiography, CT and X-ray. If the patient is on life support (ventilator, ECMO, haemodiafiltration) — changes of the settings and their lack of effect.

Q5. Where does that information come from: monitors, observation charts, the hospital information system, lab results, something else?

Как палатный и реанимационных больных есть регламент времени: реанимация - ежечасно, палатные - интервалом 4 или 8 часов, стабильные 12 часов

Оценка гемодинамики: артериальное давление, пульс, сатурация, чистота дыхание, температура; основном на мониторе или в листе назначение.

Translation: For ward patients and ICU patients there is a fixed schedule: ICU — every hour; ward patients — every 4 or 8 hours; stable patients — every 12 hours. Assessment of the haemodynamics: blood pressure, pulse, saturation, respiratory rate, temperature; mostly on the monitor or in the prescription sheet.

Q6. In practice, how is it usually noticed that a patient is getting worse? Who notices, and what happens next?

Контроль состояния пациента в ответственности врача и медсестры(брата),

В первую очередь медсестра но это зависит еще от стажа персонала.

Уведомляет врача об изменении показателей и по выше указанным данным осуществлять действия по улучшению состояния организма.

Translation: Monitoring the patient's condition is the responsibility of the doctor and the nurse. First of all the nurse, but this also depends on the staff's experience. [The nurse] notifies the doctor about changes in the measurements, and based on the data above, actions are taken to improve the patient's condition.

Q7. Where does this process work poorly or fail? If you can, give an example without patient details.

Это обучение персонала. Знание лекарственного препарата, дозировку и их действия(значение дозировок есть разница 5 тыс. ЕД гепарина и 5 мл гепарина: струйно ведение калия, что дает остановку сердце; медицинской оборудование и их механизм действии(был случай пациент на операции на аппарате ИВЛ, аппарат дал сбой и отключился, об этом узнали только когда сатурация упало: ожог пациента когда используется когуляционный аппарат нужна масса и прикосновение металом не должно быть)

Не знание нормах.

Отсутствие системы красных флагов! Кроме синий код и т.д.

Translation: It is staff training. Knowledge of a drug, its dose and its action (doses matter: there is a difference between 5,000 units of heparin and 5 ml of heparin; a bolus injection of potassium, which stops the heart); medical equipment and how it works (there was a case where a patient in surgery was on a ventilator, the device failed and switched off, and this was noticed only when the saturation dropped; a patient burn when an electrocoagulation device is used — grounding is needed and there must be no contact with metal). Not knowing the norms. The absence of a red-flag system! Apart from code blue and the like.

Q8. Which of these are used where you work or had placements?

qSOFA, SOFA, SIRS criteria / Критерии SIRS

Q9. Do such scores and alerts help in practice? Why or why not?

Шкалы безусловно помогает, но каждый пациент индивидуален

Сравнительно с одинаковым состоянием 2 пациента

У одно летальный исход через день

У второго через пять

Рассчитываться возраст, сопутствующие болезни и другие факторы

Translation: The scores certainly help, but every patient is individual. Compare two patients in the same condition: one dies after a day, the other after five. Age, comorbidities and other factors have to be taken into account.

Q10. Have you seen warnings or alerts being ignored? If so, why does that happen?

Оповещение не игнорируется!

Translation: Alerts are not ignored!

Q11. Imagine a system shows a risk-of-deterioration score for a patient. What would need to be shown next to that number for you to understand and trust it?

При правильном ведение данных пациента показывает риск ухудшение это и являются истинной, но плюс клиника и динамика

Translation: When the patient's data are recorded correctly, [the system] shows the risk of deterioration and that is true, but in addition the clinical picture and the trend [are needed].

Q12. Which of these would be useful to you?

Charts of measurements over time / Графики показателей во времени, Which measurements influenced the risk / Какие показатели повлияли на риск, A warning when data are missing or out of date / Предупреждение, если данных мало или они устарели, A limit on how often alerts repeat /

Ограничение частоты повторных оповещений, A list of patients sorted by risk / Список пациентов, отсортированный по риску

Q13. What would make you distrust such a system or avoid using it?

Такого не было.

Но шкалы очень обобщение и стандартизованные, референсные значение не учитывать индивидуальность пациента это делает врач

Translation: That has not happened. But the scores are very generalised and standardised; reference values do not take the patient's individuality into account — the doctor does that.

Is there anything important I did not ask?

Их нет

Translation: There is nothing.

Follow-up in a messenger, 21 Oct 2026, 17:26

1. Через сколько часов без новых измерений (пульс, давление, температура, дыхание, сатурация) ты бы считал(а) данные пациента устаревшими?

After how many hours without new measurements (pulse, blood pressure, temperature, breathing, saturation) would you consider a patient's data out of date?

Если не каждый час устаревшие но есть нюансы

Состояние тяжести пациента

На ИВЛ ли?

На операции?

Translation: If not every hour, they are out of date, but there are nuances: how severe the patient's condition is; is the patient on a ventilator? in surgery?

2. Как часто можно повторять оповещение по одному и тому же пациенту, чтобы оно не начало раздражать?

How often can an alert be repeated for the same patient before it starts to annoy?

Каждое изменение

Translation: At every change.

3. Что понятнее при оценке риска: число (например, 0.7), цвет (зелёный / жёлтый / красный) или слово (низкий / высокий)?

What is clearer for assessing risk: a number (for example 0.7), a colour (green / yellow / red) or a word (low / high)?

Для меня процентное соотношение

Translation: For me, a percentage.

4. Вспомни случай, когда ухудшение заметили поздно (без данных пациента). Что помешало заметить раньше?

Think of a case when deterioration was noticed late (without patient details). What prevented noticing it earlier?

Большой количество пациентов, ночные время

Translation: A large number of patients, night time.

5. Каких показателей не хватает на экране, кроме пульса, температуры, дыхания, давления и сатурации?

Which indicators are missing on the screen, apart from pulse, temperature, breathing, blood pressure and saturation?

Аппараты мониторинга всеми оснащены нет лишнего

Translation: The monitoring devices are equipped with everything; nothing is superfluous.

Appendix B. Traceability of requirements

Each Must and Should requirement can be followed from its use case to the component that implements it and the test that verifies it. Test identifiers: UT – unit test, AT – acceptance test, IN – inspection, DM – demonstration.

Requirement	Use case	Component	Verified by
FR-01 Import and validate	UC-05	Data validation, ingestion	UT-01, AT-01
FR-02 No future data	UC-06	Feature builder	UT-02
FR-03 One-command training	UC-06	Training, model store	AT-03
FR-04 Evaluation report	UC-07	Evaluation	AT-04
FR-05 Partial SIRS	UC-06	Training	UT-05
FR-06 Prediction API	UC-09	Prediction router, inference service	AT-06
FR-07 Alerts, suppression, re-alert	UC-04, UC-08	Alert logic	UT-07
FR-08 Patient list	UC-01	Patient list view, patients router	AT-08
FR-09 Trends and time since measurement	UC-02	Patient detail view, patients router	AT-09
FR-10 Contributing measurements	UC-03	Explanation, inference service	AT-10
FR-11 Insufficient data	UC-04, UC-08	Alert logic	UT-11
FR-12 Stored predictions	UC-01, UC-02	Prediction router, database	AT-12
FR-13 Model comparison view	UC-07	Comparison view, models router	AT-13
FR-14 Model metadata	UC-09	Models router	AT-14
FR-19 Fixed prompt	UC-04	Patient detail view	AT-19
FR-20 Acknowledgement	UC-10, UC-11	Alerts router, scheduler, patient detail view	UT-20, AT-20
FR-21 Data source and hourly cycle	UC-05	Data-source interface, replay adapter, scheduler	UT-21, AT-21
NFR-01 Reproducibility	UC-06, UC-07	Training, evaluation, model store	AT-N01
NFR-02 Data protection	—	Repository rules	IN-N02
NFR-03 Performance	UC-09	Prediction router, inference service	AT-N03
NFR-04 Test coverage	—	Core and backend	IN-N04
NFR-05 Portability	—	Whole system, README	DM-N05
NFR-06 Safe interpretation	UC-02	Dashboard, prediction router	UT-N06, IN-N06
NFR-07 Usability	UC-02, UC-03	Patient detail view	UX-N07
NFR-08 Transparency	—	Model card	IN-N08
NFR-09 Robust errors	UC-09	API routers	UT-N09
NFR-10 Audit records	UC-10	Alerts router, database	AT-N10